

CALIFORNIA STATE UNIVERSITY, NORTHRIDGE

Leveraging Predictive Analytics and Social Network Analysis for Enhanced
Student Performance in Education

A Thesis submitted in partial fulfillment of the requirements for the degree of
Master of Science in Computer Science

By

Sandya Bethina

December - 2025

© 2025

Sandya Bethina

ALL RIGHTS RESERVED

The thesis of Sandya Bethina is approved:

Dr. McIlhenny, Robert D, Ph.D.

Date

Dr. Ebrahimi, Mahdi, Ph.D.

Date

Dr. Wang, Taehyung, Ph.D., Chair

Date

California State University, Northridge

Acknowledgment

The present is my unstinting dedication to Professor Dr. Wang, Taehyung, who has provided a guiding hand that has radically transformed this piece of research work. Their thoughts and knowledge served to influence this research paper and unlocked a number of approaches when it comes to unveiling my method of predicting student performance using predictive analytic. Their positive feedback and considerate discussions a great jump in the quality of this work.

I would also like to express my unerasable gratitude to California State University, Northridge, which was a great place to study, and all the materials needed to conduct this research. The available literature, computers and technical support were also useful when writing this study. I would like to recognize the efforts of the faculty and administrative support staff that provided further assistance and created a learning ecosystem to promote innovation and research.

I wish to thank you all heartedly to my peers and colleagues who have proven to be a great source of motivation and inspiration. All the helpful tips that I have gained and shared dialogues all lead to me being an excellent source of residing in this research. And lastly, thank you to my family and friends who have consistently given me their fullest support and encouragement, and who believed in me and pushed me to even higher standards. The outcome of this research is the concern of every one, who at one time or other supported me in this and as much as the research acknowledges all great contribution made in that way.

Dedication

I would, with my whole heart, like to thank, in every sense, Professor Dr. Wang, Taehyung, to the very last of this journey of research, the invaluable direction, sustenance, and encouragement he has given me. Their experience and knowledge have been invaluable in this research and helped me to experiment with different methodologies and improve my method of forecasting student achievement using predictive analytics. Critical commentary and considerate deliberation has greatly enhanced this quality work.

I am indebted to the California State University, Northridge, which as I contend, has given me one of the best academic environments and resources, to allow me to conduct research at that level. Availability of applicable literature, calculating software and technical assistance have played a critical role in ensuring that this research study is completed successfully. I am grateful to the members of the faculty who have continuously supported my institutes and ensured that they are a fantastic learning environment of innovation and research.

I am quite thankful that my colleagues are highly motivational and inspirational. All this advantage has been passed on to my research by their precious recommendations, discussions, and their readiness to provide sufficient information. Final but not the least to my family and friends, as they provided a great deal of encouragement and support throughout this journey. Without their faith in me, I would have broke down in the difficult times. So it can indeed be stated that such a research was crowned success and this was made such by everyone who supported me and I owe my credit to them and my project was a success.

Table of Contents

Copyright Page.....	II
Signatures Page	III
Acknowledgment	IV
Dedication.....	V
Table of Contents	VI
List of Figures	VIII
List of Abbreviations	IX
Abstract.....	1
Chapter 1 - Introduction.....	2
1.1 Background of the Study.....	2
1.2 Motivation and Challenges in Student Performance Prediction	3
1.2.1 Motivation	3
1.2.2 Challenges	4
1.3 Problem Statement.....	5
1.4 Objectives of the Study	6
1.5 Scope of the Research	7
1.6 Significance of the Study	7
1.7 Research Questions.....	8
1.8 Research Methodology Overview	9
1.9 Contribution of the Study	11
Chapter 2 – Literature Survey	12
2.1 Introduction to Predictive Analytics in Education.....	12
2.2 Machine Learning Models in Education	13
2.3 Federated Learning in Education.....	13
2.4 Deep Learning in Student Performance Prediction.....	14
2.5 Role of Social Network Analysis (SNA)	15
2.6 Explainable AI (XAI) in Education	16
2.7 Gaps in Existing Research.....	17
2.8 Summary of Findings.....	18
Chapter 3 – Research Methodology	20
3.1 Research Design	20
3.2 Dataset Description.....	20
3.3 Feature Selection and Engineering	22
3.4 Model Selection and Justification.....	23
3.5 Explainability with XAI	25
3.6 Evaluation Metrics	26
3.7 Experimental Setup.....	28
Chapter 4 – Implementation.....	29

4.1 Dataset Description.....	29
4.2 Feature Description.....	29
4.3. Exploratory Data Analysis (EDA).....	31
4.3.1 Information About the Data.....	32
4.3.2 Statistical Description of Numerical Data	33
4.3.3 Gender Distribution.....	34
4.3.4 Distribution of Grades.....	35
4.3.5 Missing Value Inspection	36
4.3.6 Distribution of Highest Education	37
4.3.7 Final Result Vs Gender Comparison.....	38
4.3.8 Final Result Vs Region Comparison.....	39
4.3.9 Pair Plot	39
4.3.10 Correlation.....	40
4.4 Logistic Regression.....	41
4.4.1 Explainability AI for Logistic Regression.....	43
4.5 Random Forest Classifier	46
4.5.1 Explainability AI for Random Forest.....	48
4.6 LDA Classifier.....	51
6.3.1 Explainability AI for LDA	52
4.7 XGBoost.....	55
4.7.1 Explainability AI for XGBoost.....	57
4.8 CNN.....	60
4.8.1 Explainability AI for CNN	62
4.9 Federated Learning	63
4.10 Social Network Analysis (SNA).....	64
Chapter 5 – Results and Discussion	69
5.1 Overview of Experimental Results	69
5.2 Individual Model Performance Analysis.....	69
5.3 Cross-Model Comparative Insights	70
5.4 Privacy and Security Considerations	71
5.5 Explainability Results.....	71
Chapter 6 – Conclusion and Future Work	73
6.1 Summary of Objectives and Achievements	73
6.2 Key Findings	73
6.3 Limitations of the Study.....	74
6.4 Future Research Directions	74
6.5 Final Remarks.....	75
References.....	77

List of Figures

Figure 1 Research Methodology.....	10
Figure 2 First 5 rows of the data	31
Figure 3 Information regarding the data	32
Figure 4 Statistical description	33
Figure 5 Gender distribution.....	35
Figure 6 Distribution of grades	35
Figure 7 Missing value inspection	36
Figure 8 Distribution of highest education.....	37
Figure 9 Final result Vs gender comparison.....	38
Figure 10 Final Result Vs Region Comparison plot	39
Figure 11 Higher Education Aspirants Achieve Better Grades.....	40
Figure 12 Correlation plot	41
Figure 13 Table differentiating the precision, F1-Score, Accuracy.....	41
Figure 14 SHAP for logistic regression	44
Figure 15 LIME for logistic regression.....	45
Figure 16 Table differentiating the Training scenario	41
Figure 17 SHAP for random forest.....	48
Figure 18 LIME for random forest	49
Figure 19 Table differentiating the metric	52
Figure 20 SHAP for LDA.....	53
Figure 21 LIME for LDA	54
Figure 22 Table differentiating the metric	54
Figure 23 SHAP for xgboost	57
Figure 24 LIME for XGBoost	58
Figure 25 Table differentiating the metric	61
Figure 26 SHAP for CNN	62
Figure 27 Federated learning accuracy per round.....	62
Figure 28 SNA graph based on attribute	66
Figure 29 Community graph.....	67
Figure 30 Node degree distribution	67

List of Abbreviations

ML – Machine Learning

DL – Deep Learning

SNA – Social Network Analysis

FL – Federated Learning

AI – Artificial Intelligence

XGBoost – Extreme Gradient Boosting

CNN – Convolutional Neural Network

RF – Random Forest

LDA – Linear Discriminant Analysis

SHAP – SHapley Additive exPlanations

LIME – Local Interpretable Model-agnostic Explanations

KNN – K-Nearest Neighbors

F1-Score – F1 Measure (used for classification model evaluation)

TPU – Tensor Processing Unit

FL – Federated Learning

Abstract

Leveraging Predictive Analytics and Social Network Analysis for Enhanced Student Performance in Education

By

Sandya Bethina

Master of Science in Computer Science

This study explores how predictive analytics may transform student learning performance outcomes with the use of data driven decision-making especially in the use of performance identification and management of student-performance. Combining both machine and deep learning concepts, including Social Network Analysis (SNA), Logistic Regression (LR), Random Forest (RF), Linear Discriminant Analysis (LDA), XGBoost, Convolutional Neural Networks (CNN), Federated Learning (FL), the present study creates a predictive system that effectively predicts the performance of students. Compared to the traditional machine learning techniques, CNN, a deep learning model, is able to excel in identifying complex patterns and relationships in academic data and, thus, it augments the solution of predicting student performance.

The federated learning (FL) plays a vital role in ensuring privacy of information and functioning of collaborative learning on decentralized student information. Using this approach, the training can proceed to give models in other institutions or schools to ensure sensitive student data is not lost. The explainable AI (XAI) deployed in all the models can assist in providing transparency and interpretability to enable educators make decisions based on logical considerations about the factors that appear to have the most significant influence on student outcomes. LR and LDA can provide and build linear decision edges but RF and XGBoost can support intricate data interactions satisfactorily. SNA, in its turn, has a crucial role in the research of peer networks and significance of collaboration in academic success. This type of research could be useful to personalised learning recommendations, early intervention practice, and data-driven education policies based on predictive analytics alongside Explainable AI and Federated Learning. Such innovative technologies used together enable the educators to enhance the performance of the students further without violating the data security and privacy.

Chapter 1 - Introduction

1.1 Background of the Study

In the contemporary world, prediction of student performance has become an important issue in many institutions and colleges to enhance academic performance. There is abundant data today in the study of learning patterns to predictive analytics, including to determine at-risk students and to implement very specific interventions. It is a feature of decision-making that the institutions desire to enhance in the learning experiences and teaching techniques with the aid of predicting performance of students.

Traditional assessments rely on post intervals assessments which may not be the most relevant in forecasting the academic achievements of either Ajay or Girish. The more recent generation more dynamic and data-oriented applications have machine learning-based and deep learning-based predictive models. They are very accurate applications in predicting the performance of students based on past academic history and other aspects which influence the attendance and correlation of a social background of students in order to assist in determining trends and patterns that influence the success of students.

Machine learning, such as Logistics Regression, Random forest, and XGBoost, will be paired with deep learning algorithms to predict student outcomes, such as CNNs, better. SNA gives an insight into peer pressure and peer cooperation in terms of academic performance. Explainable AI ensures the predictive model was interpretable and transparent to ensure the users/educators can see how the decisions were made.

Therefore, schools are progressively shifting to these sophisticated technologies towards the progressive future in education. Their assistance is useful to recognise students requiring immediate assistance and provide rapid intervention and personalized aid, as well as curriculum adjustments that establish a facilitating environment to achieve higher performance. Predictive analytics in education is depicted to be highly relevant and highly likely to change the definition and the experience of student performance towards improved learning.

1.2 Motivation and Challenges in Student Performance Prediction

1.2.1 Motivation

The need to have high and customized educational experience has necessitated high performance predictive model of student performance. Educational establishments, educators and policy makers are working towards trying to ensure that the performance of students improves by finding out all the aspects that can influence the success of the learners in their studies and then taking all the necessary measures to help them. The traditional performance assessment, such as examinations and regular assessment, fails to give the comprehensive measures of the numerous variables that interfere with the life and the learning process of a student. This creates the need of data-driven insights through predictive analytics, which would evaluate the academic progress and give early alerts to struggling students. Among the major driving forces in student performance prediction is the early intervention. Students experience numerous issues, such as learning difficulties, lack of motivation and external socioeconomic influences that can be used against academic performance. The early identification of at-risk students will help the institutions to receive individualized intervention like additional tutoring sessions, mentorship programs, or various other alternative interventions to enhance learning results.

Other than that, personalized learning experience is the other powerful motivator. Predictive analytics would give various educational materials and pedagogical approaches based on the personal needs of learners. Various student and past academic performance patterns would then lead to different individual learning experiences serving different student and past academic performance and therefore would help in ensuring that student potential is maximized.

In addition, predictive analytics in education would also help in the prudent optimization of educational resources. A realization of the need of students in terms of additional support will be the result of understanding that will allow allocating resources in a more cost-effective way. Predictive models are then used as a means of arriving at a verdict about efficacy in education either in the addition of supplementary teachers instructing on topics in which students generally perform the worst or using an online learning resource pushing the envelope a step higher.

1.2.2 Challenges

There are a number of problems that must be addressed in order to facilitate the implementation of more student achievement prediction.

Data quality and availability- high-quality data that is designed based on student needs is the most crucial challenge of predictive analytics. A great deal of education data might be incomplete, inconsistent or biased. Losses in the records, erroneous data entries or absence of comprehensive datasets is bias on the research outcome and pure, standardized and representative data is the basis of creating reliable constructs.

- Feature Selection and Interpretability: It is not straightforward to find out what factors are relevant to the performance of students. As an illustration, academic records, attendance, social behavior, and psychological behavior translate to the performance of students. The terms that are the most relevant should be selected and made friendly to interpret. Since most machine learning models, in particular, deep learning models such as CNNs are black boxes, the model is not consulted to explain why a specific prediction was made. Explainable AI (XAI) can resolve this problem.
- Bias and Fairness in Prediction: Predictive model can be biased. It is frequently using non diversified and non representative training data. The influence of other demographic factors such as gender, ethnicity, and other socio-economic factors may occasionally result in prediction bias, hence an unequal prediction. The development of ethical AI and bias reduction strategies to resolve bias is highly necessary as a way of making predictions not a perpetuation of inequities in the education sector.
- Dynamic learning environments: The learning behavior of a students is dynamic. It is subject to change. The performance can be influenced by the new teaching approach that varies, personal factors, etc. The purpose or the desire alteration. A model is trained using past data and that will not necessarily model well in future situations. The models will have to undergo continuous learning and updating to guarantee further accuracy in prediction.
- Privacy and Security Concern: Student data is highly sensitive and that is where this becomes a big dilemma when it comes to privacy and security. An illegal access and data breach would probably have ethical and legal implications. Thus, the education-related protection implementation along with compliance with privacy regulations is obligatory when working with the data.
- Adoption Resistance: In this aspect, many institutions continue to believe in the old ways

of assessment and might be reluctant to adopt predictive models that are driven by artificial intelligence. The probable perception and confidence about machine learning predictions may be generally deficient in the minds of teachers and administrators. In order to take the process of predictive analytics adoption further, appropriate training and trust building and demonstrating the value of the predictive analytics will become the part of the process.

- Computational Complexity and Cost: Newer, modern machine learning and deep learning models are greedy on the machine in terms of the machine resources. Schools and universities with tight budgets may be able to afford very little of such machines, hence making a differentiation by lack of resources.

1.3 Problem Statement

The issue of evaluation of student performance is very complicated in the world of educational institutions. Conventional forms of assessment like examinations, quizzes, and assessments aim to give a constrained view of the academic growth of a student. Regrettably, insufficient emphasis is placed on concomitants that are more extensively related to the process of learning and include such aspects of learning as attendance, participation, socio-economic condition, or peer interaction. Hence, the process in which the educator takes actions to mitigate a situation once he/she has identified a poor-performing student is quite late to make a decision that can change anything.

The main issue when it comes to measuring the performance of students is the unavailability of real-time information. Majority of the institutions end up using end of semester assessment methods, making them ineffective in terms of timely assistance or remedial action. Lateness of recognition of underachieving students is detrimental to retention and academic achievement. In addition, the large volumes of data produced by students make manual assessment impossible, which results in inconsistent assessments.

The way out of these limitations is predictive analytics, which is a method of assessing students based on data. Academic records, behavioral tendencies, and social interactions are analyzed with the help of machine learning and deep learning models to determine the future success of students in the institution. The effectiveness of the entire prediction processes is improved by some models used to predict analytics, such as Logistic Regression, Random Forest, XGBoost, and CNN. This is accomplished by Transparency and Explainable AI (XAI) which is where the educator can learn how the prediction was achieved.

Therefore, predictive analytics in education is a necessity because it would enable the

implementation of interventions at an early stage, customize learning, and aid decision-making. Predictive analytics will enable institutions to accommodate students, coordinate resources essential in the learning process and enhance academic achievements.

1.4 Objectives of the Study

The main focus of this research is to construct an accurate and explainable predictive model of the performance of students. This research is expected to create a machine learning- and deep learning-based data-driven intervention capable of assisting educational institutions in identifying students at risk and introduce timely interventions to facilitate learning.

Another crucial aspect of this research, model development, uses various algorithms such as Logistic Regression (LR), Random Forest (RF), XGBoost and Convolutional neural networks (CNN). The model will examine different academic and behavioral data to predict the success or threat of underperformance of students. In addition, we will use Social Network Analysis (SNA) to observe the impact of peer interaction on the learning outcome of a student.

Another significance of importance is interpretability. In the majority of cases, machine learning models, particularly those that are grounded in deep learning, operate as black boxes that teachers can hardly comprehend why some predictions are made. Such will be overcome with Explainable AI (XAI) methods integrated into all models to provide understandable, albeit not terribly actionable, insights. This will ensure transparency, trust and easy interpretation between the educators and the decision-makers.

Lastly, early intervention will be encouraged by this study. By anticipating the performance of students in advance, teachers can take proactive steps including specific tutoring, individualized mentoring and curriculum modification of underperforming students. This research will eventually streamline the learning materials by making the intervention focus on the learners who require it the most. The general idea is to fill the gap existing between traditional forms of assessment and the up-to-date data based methods by creating a predictive system that is interpretable in making decisions informed and on time on how to improve student success.

1.5 Scope of the Research

The current study observes the use of machine learning and deep learning models of

predictive analytics in determining student performance. The concept is to be able to pinpoint player variables that can potentially correlate with academic achievement and form a model that can be interpreted to provide correct predictions. The intended study is contextualized in terms of the datasets, models and evaluation criteria used in the study. The student performance datasets would maintain details of academics, attendance, behavior, and interaction of the student with peers, and the preprocessing of the datasets would eliminate inconsistencies in the information which would guarantee the quality of the data. The research, however, does not look into the real time data collection configurations but uses past data to train and test their models.

The analysis involves application of a hybrid model of machine learning and deep learning coupled with Logistic Regression (LR), Random Forest (RF), XGBoost and Convolutional Neural Networks (CNN). The analyze the effects of peer collaboration will also be part of the role in Social Network Analysis (SNA). Which adopts explainable AI (XAI) techniques that further contribute to the improved interpretation of the models, allowing predicting in a way that is understandable and usable. The evaluation criteria here is accuracy, precision, recall and F1 score. Hardware acceleration and distributed computing were not discussed in the study, but the attention was paid to the modifications of algorithms and their features selection to achieve higher results. The study will set out to develop a useful prediction framework that can be implemented in the education sector once limited, hence inviting the institutions to apply it in their decision-making to enhance student performance. Results will subsequently be embraced to implement early intervention measures and individualized learning endeavors to enhance academic excellence.

1.6 Significance of the Study

Predictive analytics can contribute a lot to transform the education landscape through its ability to implement an empirical decision-making process, operating in both directions between students and educators. Traditionally, the majority of evaluation techniques have been applied occasionally and they provide at most of a snapshot of how the students have performed over time; therefore, solutions via predictive model paths could enable institutions to take preventive measures to identify risk students before the occurrence of the potential impairment and prescribe remedial and individualized assistance.

Due to predictive analytics, students receive more personal learning. The set of scholastic records, attendance, and behaviors act as forecasts of learning disparities which

could be recovered using personalized approach where the students are motivated to accept the assistance when they are still performing well before their performance are prone to tumble. This enhances retention rates and the general performance in their education. To some degree, predictive analytics can be used to give indications about the classroom and learning behavior patterns of students. These insights would enable teachers to adjust their teaching strategies, profile students who need additional help and to develop targeted intervention plans. The ease with which educators can turn to and consider the results is because such predictions can be explained using XAI.

This also implies that institutions will be in a position to make adjustments in their resource distributions in relation to subjects or areas in which the students seem to be facing the most challenges. This may be achieved by specific tutoring packages, mentoring, and other interventions like curriculum reform. This paper demonstrates that predictive analytics may help bridge the gap between the traditional rule-book learning and the current data-driven learning, making it more flexible, effective, and student-centric. Such kind of pro-active measures would result in marked changes in the education outcomes of the institution across the board.

1.7 Research Questions

The study is associated with a large number of relevant research questions that will facilitate its completion along with the performance of predictive analytic tools in terms of student performance. It also will introduce superior knowledge, with these, regarding how machine learning and deep learning models may enhance the results of education.

- What is the effectiveness of machine learning and deep learning models in predicting student performance?

This concerns the tested precision and dependability of standard models such as the Logistic Regression (LR) model, the Random Forest (RF) model, the XGBoost, and the Convolutional Neural Networks (CNN) when studying the student data and prediction.

- what are the most predictive variables that determine student success?

This study will determine the major predictors that trial success by determining the most important factors that are associated with academic performance and behavioral factors and social characteristics that influence learning outcomes.

- What can explain the role of Social Network Analysis (SNA) in the academic success of peers?

This paper examines the impact of student interactions, collaborations and relationship with peers on the learning process and performance.

- What does Explainable AI do to enhance the XAIinterpretability of the predictive model to students?

It would also examine how the black box properties of different machine learning-centric models can be lifted with the help of XAI methods in order to make the predictions easier to interpret by teachers.

- Predictive analytics in facilitating early intervention plans as well as individual learning?

This query raises the issue of how a learning institution can utilise the predictive knowledge to inform the intervention methods presented to the students who are struggling with the methods of teaching.

In this way, by responding to these questions, this study will develop a fully functioning model of a robust predictive framework that would enhance educational decision-making and improved student outcomes.

1.8 Research Methodology Overview

The foundation of data-driven methodologies applied to the study is student performance prediction using machine learning and deep learning models. The suggested methodology will have various steps, where one will collect data, preprocess data, develop a model, assess it, and explain it with the help of Explainable AI or XAI.

Data collection, the first most important agenda, entails data sets of students, such as, academic records, attendance, behavioral patterns, and social interactions. The research is preoccupied with publicly available data-or institutional data- to meet the ethical concerns and a compliance with privacy.

Then the dataset is taken through preprocessing, cleaning and preparation. In particular, missing data treatment, numerical data normalisation, categorical data encoding features and identification of relevant variables to be analysed are the key tasks. However, it is possible to use feature engineering because it can assist the model performance by extracting some valuable knowledge using raw data. A training of different machine and deep learning models in the model development stage has so far involved Logistic Regression (LR), Random Forest (RF), XGBoost, and Convolutional Neural Networks (CNN) training. Also, there is the use of Social Network Analysis (SNA), to examine the peer influence on student performance.

Explainable AI (XAI) techniques were also integrated to maintain the interpretability of the model whereby educators could have an opportunity to interpret what they were being predicted and the key factors that impacted the outcome. The research intends to determine its model performance in terms of accuracy, precision, recall, and F1-score as indicators of performance.

Lastly, the results are analyzed with an attempt to identify trends and patterns that will guide the potential interventions in terms of early intervention or individualized learning. The general methodology is a guarantee of a thorough and systematic framework of predicting student performance, therefore, allowing institutions to participate in data-driven contribution in the field of education.

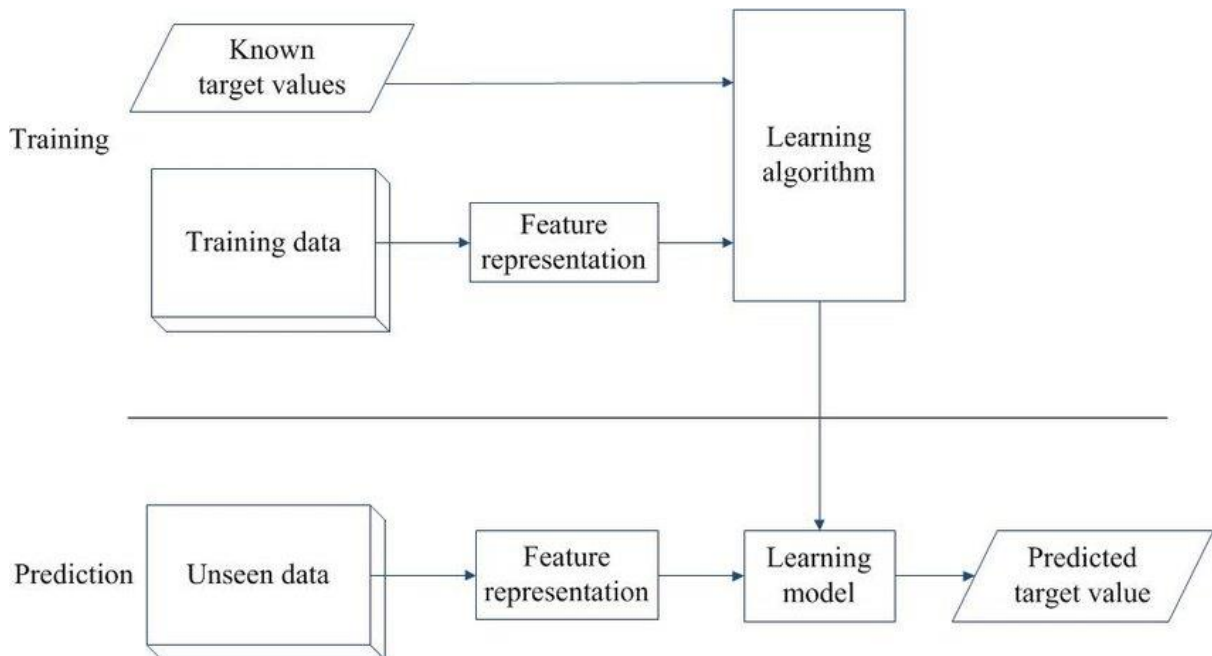


Figure 1: Research Methodology

1.9 Contribution of the Study

This work is very valuable in education because it integrates predictive analytics in evaluation of student performance. A data-driven methodology is developed through several machine learning and deep learning approaches, which are used to detect students at risk of not performing well in their studies and therefore requiring a personalized approach. The core of

the contributions is a highly advanced predictive model of student performance prediction that is thoroughly trained on Logistic Regression (LR), Random Forest (RF), XGBoost, Convolutional Neural Networks (CNN) and Federated Learning (FL). The credit provided to Social Network Analysis (SNA) provided the framework with a strong understanding of mediation processes according to which peer interaction and academic success can be related to each other.

The other contribution is about the significance of XAI techniques to enhance model transparency. The majority of the predictive models are black boxes and that is why the results can hardly be interpreted by the teachers. Therefore, the explainability inclusion in the framework will provide the generated predictions in the study to be interpretable, actionable, and trustworthy to make educators take indispensable decisions based on the information provided in these models.

Moreover, the research provides a predictive model of early intervention in education such that the school could detect troubled students before they start showing negative performance. It is through this form of intervention that the institutions are able to employ the appropriate intervention interventions such as tailored tutoring, mentorship plans, and curriculum adjustment to improve the performance of the students.

The current research paper seals a gap on the educational measurement landscape in relation to the conventional assessment paradigm and predictive analytics. It introduces a solution, which is scalable and applicable to the targeting of student learning experiences by evidence-based decisions.

Chapter 2 - Literature Survey

2.1 Introduction to Predictive Analytics in Education

The application of predictive analytics in the education spectrum has greatly become common, where the use of analytics has transformed the manner in which an institution analyzes and supports student performance. Predictive analytics is based on past data and the use of machine learning and statistical models, can help educators analyze at-risk students, understanding their learning behavior, and assist them with interventions. This dramatic utilization of the data enables institutions with decisions related to the development of student-serving activities with information on academic achievement and retention [1].

Types of students Performance prediction Predictive analytics have been widely written about in the literature. Up to this point the older ways have used a rudimentary form of statistical analysis on variables like grades of students and attendance. Recent developments in machine learning and deep-learning have observed the use of more advanced models (Logistic Regression (LR), Random Forest (RF), Support Vector Machines (SVM), XGBoost, and Convolutional Neural Networks (CNN)) to provide the best prediction accuracy, considering various variables (academic history, behavioral patterns, social interaction, and external influences-to predict success or failure) among students [2].

It was also established through research efforts that Social Network Analysis (SNA) is primary in the comprehension of peer interaction that relates to academic performance. The researches reveal that the students with a more robust academic network with their colleagues are performing better, thus also attributing the social aspects significance to learning outcomes [3]. Explainable AI (XAI), conversely, is a trend that is beginning to gain traction in the educational sector as a remedy to the interpretability obstacle to predictive algorithms. Sufficient emphasis is placed in much of the research so that adequate emphasis should be put on AI solutions that are interpretable and transparent, e.g. enabling teachers to trust and use predictions. Based on the conducted research, it is possible to draw the conclusion that predictive analytics in conjunction with machine learning, deep learning, and XAI is a possible direction toward improvement in the education field. The issues still making their way round these would be data quality, bias and privacy requirements. This work expands on the previous study because it offers a more interpretative depiction of the predictive framework of assessing the performance of students.

2.2 Machine Learning Models in Education

The edification of machine learning (ML) models has become a truth that can provide a great improvement in the analysis and prediction of the performance of students based on the perspective of education. The evaluation of academic success was done using different algorithms that involved Logistic Regression (LR), Random Forest (RF), Linear Discriminant Analysis (LDA), and XGBoost using student data including grades, attendance, behavioral pattern, and social interaction [4]. Logistic regression can be merely referred to as one of the simplest but valid forms of discrimination method to a binary classification problem such as on whether a specific student passed or failed an examination. It attempts to correlate independent variables (characteristics of students) to a dependent binary variable. LR gives interpretable coefficients and, therefore, it is used in education to help educators discover the important factors affecting performance. On the downside, it is linear. As such it might fail to emulate some intricate trends in predictor variables on students [5].

In Random Forest, a certain amount of those decision trees in an ensemble setting is built, and the predicted outputs of each tree are combined in an attempt to significantly improve predictive accuracy. Random Forests are practically very powerful when it comes to handling non-linearity and high-dimensional datasets, enabling its use in prediction of student performance. It gives a score on the importance of features, thus assisting the educators in identifying the more influential factors affecting the learning outcomes. The complexity presents another challenge with the random forest as it may be difficult to explain its prediction.

LDA is a classification method that finds the linear combination of features that characterizes or separates two or more classes. In the analysis of academic performance, LDA helps distinguish among the categories of students such as good and bad based on the features defined in a structure. This computational efficiency may be an added advantage, but linearity may remain a drawback to LDA when faced with large non-linear relationships [6].

XGBoost is an adaptive boosting algorithm that works with the gradient descent method, having a reputation for high accuracy and efficiency. Its application in education is profound for trend analysis and early detection of at-risk students. It can address missing values, deal with complex relationships, and manage large datasets since it holds advantages over traditional methods. Overfitting remains an issue, thus requiring careful tuning. By applying these ML models, institutions can create data-driven initiatives to develop student learning, personalized assistance, and above all, improved educational impacts.

2.3 Federated Learning in Education

Federated Learning (FL) represents an innovative machine learning strategy to train models that collaborate with decentralized data sources whilst maintaining data privacy. With respect to education, FL enables learning institutions to develop predictive models on student performance without having to share sensitive student records across schools, districts, or nations. Rather than centralising data, the local model of each institution is trained independently on its own data and only updates in the model (gradients) are shared with a central server. Through this collaborative paradigm, more accurate models are developed with no compromise of privacy because of unified information that obtains the knowledge of many data sources and is merged.

Discover Education FL can enhance the effectiveness of learning in educational institutions where they are able to create models that can help predict the ability of students, alert scholars who are in danger of failure, and determine individual learning pathways. It enables schools to cooperate without sharing the sensitive data about students and it assures that the privacy regulation such as GDPR or FERPA is observed. Also, the practice of FL facilitates the development of more generalized and more solid models with the inclusion of numerous educational realities, including divergent teaching strategies to socio-economic levels.

Through the assistance of FL, educators can utilize data and make robust judgments, provide early interventions and personal learning experience without violating student confidentiality. This technology also forms the basis of developing data-driven education policy that may be modified and adjusted to the variable needs of all students in the diverse regions.

2.4 Deep Learning in Student Performance Prediction

Deep learning has invaded the world of education as an effort to follow up, evaluate and anticipate complicated patterns of student information. In contrast to classical machine learning algorithms that use handcrafted features to make predictive tasks, deep-learning algorithms like Convolutional Neural Networks (CNNs) learn meaningful representations of structured data to make accurate predictive decisions.

Although CNNs are mostly known to process images, there is an application of CNNs that can predict student performance without taking the so-called traditional route. In this instance, CNN is used to reveal more complex relationships between organised numerical data, as opposed to images: The hierarchical character of learning in CNNs enables it to learn latent trends in student records, such as grading, attendance, behavioral history, and even engagement trends [7]. The fact that the CNN feature extraction process enables performance prediction dynamic models to be more adaptive and robust. These are the models that detect at-risk students. Another benefit of using CNNs in education over alternative methods is that large datasets do not need significant pretreatment to undergo the processing. Other preprocess can also refine features and thus does not have the learning process inherent in CNNs.

For instance, traditional models build statistical relationships through Logistic Regression (LR), Random Forest (RF), and XGBoost, making them unfit for adapting to changing scenarios. In contrast, CNNs make the adjustment of the learning parameters dynamically, thus facilitating complex student performance prediction tasks. Coupled with Explainable AI (XAI), CNN-based prediction models can also offer an understandable interpretation to educators, compelling them to trust and understand the predictions.

Thus, with CNN ability to learn deep representations, predictive analytics' strength in education lies in the institution making data-driven decisions to improve student outcomes.

2.5 Role of Social Network Analysis (SNA)

SNA works on the principle that interactions between peers affect student performance. While traditional models of academic performance concentrate primarily on individual characteristics such as grade point average, attendance history, or personal learning behaviors, they often ignore the impact of social connections on education. Social network analysis helps bridge this gap by analyzing student relations while highlighting collaboration, communication, and engagement patterns that significantly affect academic success. SNA treats students as nodes and their interactions as edges, representing the influence students exert on one another in a learning context- nothing new has been created here. SNA will help identify the main influencers, active learning syndicates, and socially isolated students needing additional support. Peer interaction is a core channel of influence on academic performance through knowledge sharing. "If students are involved in class discussion and group project activities and help their classmates, they probably do better." Running parallel with designing their learning environment is to use SNA to study the possible networks of the collaborative

learning environment and its effect on academic success [8].

The second element of interest is peer effect where performance of a student can be influenced by his or her nearest counterparts. Prosperous students will provide motivating conditions that will even encourage others to perform and poor peer influence may further affect academic performance. Moreover, involvement in learning process and interaction is a key element of learning. SNA can also be used to identify less engaged students in order to take precautionary actions to promote inclusivity and participation by the institutions. As SNA is included in the set of predictors of student performance available, matured educational organizations can get an insight of the varying levels of social interactions, leading to a better design of intervention strategies of potential improvement with respect to academic performance [9].

2.6 Explainable AI (XAI) in Education

As more artificial intelligence (AI) is introduced into the formal educational setting, Explainable AI (XAI) is likely to provide equal transparency and trust in predictive models. Data are traditionally modeled by machine learning and deep learning techniques in many ways: examples include Random Forest (RF), XGBoost, or even Convolutional Neural Networks (CNN)-where they are largely a black box, and little can be known about how they make predictions. Therefore, the breakdown of trust or the matters related to trust in such AI-derived insights can impede such tools implementation in the academic community both by the educators and decision-makers [10].

In fact, XAI can offer the answers because it is a way of showing how the models arrive at their forecast, allowing an educator to know what would have contributed to the student performance. As an illustration, through the data on major variables, such as attendance, participation, and involvement of social engagement, it enables the institutions to be fair and make decisions to build decisions on the basis of data and be accountable in their evaluations. It will shortly assist to exclude possible biases in the predictive model ensuring that no learner gets into a misclassification based on non-relevant or prejudiced characteristics [11].

Readable AI models will enable teachers and administrators to become more intervening agents in the initial stages, as they will identify the at-risk students and create learning paths to them. They also assist students in gaining clearer insights concerning their

weaknesses and strengths and thus the learning process becomes personal and visible to the learners [12].

Thus, by incorporating the XAI into predictive analytics, it will be possible to reduce the gap between the high potentials of AI technologies and their practical implementation in the real world, making sure that the decisions made with the help of the AI system are both trustworthy, moral, and meaningful. This process eventually enhances the quality of student learning experiences as well as resulting to equity in and efficiency of the education system [13].

2.7 Gaps in Existing Research

Although dramatic progress has been made in predictive analytics in the field of education, glaring gaps exist in the existing literature concerning its applicability in models of student performance prediction. One of the greatest drawbacks of the vast majority of earlier studies can be explained by a lack of datasets which can have sufficient description. Majority of them have plunged in only to come up with grades, attendance, and limited behavioral and psychological indices with no efforts to appreciate the social aspects that play a pivotal role in determining the performance of students. This decreases the effectiveness and external validity of the models under consideration [14].

A second problem is the conventionality bias as studies over-utilized known machine learning models that include Logistic Regression, Decision Trees, and Random Forest. These are useful models, but their use has frequently proved inefficient in dealing with certain difficult, non-linear correlations between the data of students. CNNs and other deep learning models have demonstrated higher accuracy in prediction, but have not been used extensively in education settings because of the barriers of interpretability and computation costs. That field of study is still young since most publications still base their assumptions on the combination of Explainable Artificial Intelligence and deep learning models to enhance their trust and transparency quotient in their performance prediction of student performances [15]. Most of the existing models also have some bias and fairness problems remaining. Although AI must not guide decisions that discriminate against groups of students based on gender, socio-economic status, or ethnicity, predictive analytics in education are largely unsuccessful in addressing most of these problems. Majority of the studies have not found any such bias and mitigation plans and that means that the ethical issues could be unaddressed.

A majority of the patterns are retrospective-oriented and are not predictive-action oriented in

real-time. The analysis of data is usually conducted on historical grounds. They do not have a live tracking system, with the help of which teachers would take proactive actions. It is likely that future research would invest in coming up with dynamic and adaptive models that are constantly updated with new inputs provided by the students [16].

To fill these gaps, an all-encompassing approach to predicting student performance would encompass the combination of all diversity data points, enhance the model interpretability, bias reduction, and implement real-time interventions with the perspective of enhancing education results.

2.8 Summary of Findings

The reviews of the literature foretell predictive analytics as constantly growing relevance to education on the basis that, probably, it can help to improve the measurement of student performance, as well as interventions. The other notable finding is that machine learning models (e.g., Logistic Regression or LR), Random Forest (RF), XGBoost, and even deep learning models (e.g., Convolutional Neural Networks or CNN) introduce additional weight to the prediction since they are cast together with the classical assessment process. These models are aimed at identifying the trends in academic and behavioral data, as a result the institution is able to initiate a proactive intervention to support students [17].

The second and perhaps equally important finding is that Social Network Analysis (SNA) can be useful in the research of academic performance at the peer level. Collaboration, working, study groups, and interaction with high-performers enhance the academic performance by far. Combining SNA and predictive analytics offers a possibility of a more detailed representation on student performance.

According to the literature, explainable AI (XAI), thus, is highly valuable in education to provide transparency of the predictive models and a sense of trust in the predictive models. Because most deep learning methods often assume black box methods, the integration of XAI will empower teachers to gain a deeper understanding of how the forecasts are achieved and, thus, allow them to implement data-driven decisions based on it [18].

Moreover, the identified gaps of the current research are the data diversity, model bias, and real-time prediction capabilities. Predictive analytics have indeed improved the prediction of student performance but there are gaps in the datasets employed by various studies since they do not include behavioral and socioeconomic variables. Sealing these will contribute to the better prediction systems of student performance.

Therefore, the current research is inspired by that finding and develops an interpretable, AI-based framework through combining social network analysis (SNA), explainable artificial intelligence (XAI), and state-of-the-art machine learning models which form a comprehensive system of predicting student success and planning interventions.

Chapter 3 – Research Methodology

3.1 Research Design

This study aims at coming up with a clear systematic approach to the creation of proper predictive models in respect to the assessment of the student performance. The research would be done in several phases- data collection, data preprocessing, model selection, training, evaluation, and interpretation using Explainable AI (XAI).

Phase one: gathers information on the performance of the student-his grades, school attendance, levels of engagement with the classroom setting, and social interactions. The source of data is open educational data or institution records, on the basis of which a representative and multi-dimensional sample can be formed. The data is then subjected to techniques of missing value treatment, normalization of numerical features and encoding categorical variables to enhance the quality of the data.

Concerning the step of model selection, different machine learning and deep learning models are selected based on their predictive potentials: Logistic Regression (LR), Random Forest (RF), Linear Discriminant Analysis (LDA), and XGBoost, because they are efficient in predicting with structured data, and simultaneously, Convolutional Neural Networks are employed to reflect the complex patterns related to student data that extend outside image processing, and Social Network Analysis includes the peer influence in the academic success.

The determinants of the evaluation phase used in measuring the model performance are accuracy, precision, recall and F1-score to determine how effective the models are. Lastly, XAI methods are a subset of the efforts to achieve transparency that would enable educating the fact that the predictions can be made. With this research design, the study will attempt to create a predictive model that is reliable and interpretable to help institutions to identify at-risk students, maximize learning plans, and enhance academic results based on data-informed choices.

3.2 Dataset Description

The OULA Dataset is primarily used for educational research and contains vital information pertaining to students' learning behavior and academic performance. It is significant since almost all information contained in this dataset is derived from online

settings, which can predict students' level of success and possible reasons for success or failure.

Characteristics of the Dataset

Different features supplement the OULA data from several databases; these different attributes of students in this dataset could be demographic, academic, behavioral, and interaction-based. Notable features include:

- Demographic Attributes: Age, Gender, educational background, and employment.
- Academic Attributes: Enrollment into courses, marks, assignments submitted, and final grades attained.
- Behavioral Attributes: Logins into the platform, time spent on learning resources, and participation in forums.
- Interaction-based attributes: peer discussions, activities, and student engagement in collaborative learning settings.

Data Source

The OULA dataset consists of data collected from online university learning platforms directed at student interaction and evaluation of the performance metrics in real-time. This suggests therefore that this dataset is suitable for studying how behaviors under digital learning aspects affect levels of success in higher education; thus, it is also very intriguing to modern educational research.

Data Preprocessing

This will necessitate a certain amount of preprocessing before the dataset is applied to predictive modeling including:

- Missing Value: Missing values are handled by imputing them with the help of statistics.
- Feature Scaling: Scaling numerical features to be equal.
- Categorical Variable Encoding: Particularly used to encode string-like data (i.e. course names) into numerical values.
- Outlier detect: To verify that the data is reliable, the entire data set is checked through outlier detection and outliers are somehow taken into consideration.

The study aims at coming up with an efficient predictive model of identifying high-risk students using the OULA data to prescribe and offer early intervention to enhance the learning process of such students.

3.3 Feature Selection and Engineering

Feature Selection and Engineering are the main guiding factors which keep close watch on achieving-the-accuracy, and interpreting predictive models for student performance analysis. Hence, the most relevant predictors should be identified in order that the model focuses on important predictors andamp avoids noise as well as computation complexity.

First Feature Selection

The very first step in feature selection must specify what variables have the most significant impact on student success. The defining features impacting students' performance based on past research and basing on the OULA dataset are:

- Demographics: Age, gender, past education status, and working status have an influence on learning behavior.
- Academic Performance Indexes: Grades, scores on assignments, performance assessments, and successful completion of courses predict future success.
- Engagements and behaviours: This will include the number of logins into the system, the average time spent on the material pertinent to the course, forum discussion, and access to learning materials- all these demonstrate the desire of the student.
- Social Network Features: The interaction with peers, group discussions and patterns of collaborative studying influence the extent to which one goes in the learning process.

Use of Correlation and feature importance measures using Mutual Information, Recursive Feature Elimination (RFE), and SHAP values, would significantly aid in finding strong predictors with minimal redundancy or irrelevance.

Feature Engineering

After the feature selection, it will use the feature engineering to derive the new meaningful representations of the data. There are tools that are of great importance:

- Aggregate Student Activity: The new aggregated logins aggregated to study will be exposed at varying times.
- Establish Trends of Performance: Rolling averages of grades and engagement levels

that tracks one performance over a time span.

- Encoding Categorical Data: It transforms categorical variables (types of course, learning styles and types of participation in discussion forums) into numbers to be model-compatible.

In such a way, by applying the most effective strategies to choose and design features, it will be adequate to bring the mission of this research study to reality; to enhance its validity and reliability in drawing predictions in the performance of students using machine learning.

3.4 Model Selection and Justification

To a great degree, the choice of predictive models is a highly significant step that must be considered when creating a system to predict successful performances of students. All machine learning and deep learning approaches reflect various views of the academic performance. Social Network Analysis-based (SNA), Logistic Regression (LR), Random Forest (RF), Linear Discriminant Analysis (LDA), XGBoost, and Convolutional Neural Networks models are some examples that continue this paper to build very broad and narrow prediction models.

Social Network Analysis (SNA) will be included to study academic performance in terms of peer influence and social interaction. As it has been noted above, the collaboration of students, their discussion forums or participation in study groups are the phenomena, which are most likely to have crucial impacts. The outcome of this analysis will aid in identifying the critical influencers, the level of student engagement, and the collaboration learning patterns which will result in successful outcomes. These effects can be explored using social relationship and will provide some positive contribution as far as peer dynamics are concerned in relation to student performance which is what this study is fundamentally dealing with.

Logistic Regression (LR) is one of the most popular and traditionally applied classification algorithms when it comes to interpreting a student success-failure predictive model as output. Since the majority of the student performance prediction cases could be presented as a binary classification task (such as the pass-fail or high-risk-low-risk categories), LR is therefore suitable. Moreover, LR is also computationally efficient and illuminating on the significance of such factors that influence the outcome of the student. With its simplicity, it can be described as a good baseline model to use.

Random Forest (RF) is an ensemble learning algorithm, or to put it more simply, model

prediction is built through trees classifiers. They are especially useful in dealing with non-linear relationships that are complex and also high dimensional data. In the example of student performance prediction, it assists in capturing the intricate interaction of documents regarding a person that discuss his/her academic situation, behavioral tendencies and participation rates. The other strength of RF is that they provide a chance to assign importance scores to different features that are regarded as a part of the interpretation process, thus opening the door to the most significant factors that influence student success.

Linear Discriminant Analysis (LDA) was selected since it assists in identifying the optimal combination of linear features that can be used to distinguish the performance groups of students. This works out to the benefit of several classification labels like high, average and struggling students. LDA is optimal when the classes are distinctly separated in the data set and is less complex to calculate than a more complicated model. This would therefore be a systematic way of forecasting the success of the students and improve their interpretability.

XGBoost is one of the best performing gradients boosting algorithms. These days, it is extensively used in predictive analytics because of its efficiency and accuracy. It is able to manage immense datasets, lack of values, and complex feature interactions, which makes it all the more suitable for predicting student performance. XGBoost improves the traditional tree-based models to avoid overfitting and improve the performance using boosting techniques. So far learning patterns from student data adaptively makes it one of the powerful models in this research.

Convolutional Neural Networks (CNN) are used mostly for image processing, but slowly, they are getting important in structured data because of their ability to learn hierarchical patterns. CNN will help us identify the complex relationship of student attributes and engagement behaviors with records of academics. CNN deep begins to capture nonlinear dependencies which traditional models may miss. Though CNN is expensive in computation, it is more accurate and adaptable for analyzing student performance trends.

Federation Learning (FL) is a type of client-server architecture, consisting of a central server that helps to coordinate the entire training process, and local clients (institutions, or few classes) that train model data on decentralized data. It is a data privacy set up, in which data is partitioned at institute or class or group levels. FL interacts with ML or DL models such as Random Forests or CNNs, that is, the peripheral devices update and communicate a set of local model parameters or equivalently, gradients, to the central server. This information is used on the server through aggregation algorithms such as Federated Averaging (FedAvg) or FedProx to improve the world model. Communication is performed in several rounds,

whereby clients provide periodical updates to the model at the server, without any sensitive data leaving the walls of the local institution but still utilizing the power of collective learning.

Each model adds a different angle to the prediction. SNA adds social context to predictions, LR and LDA add interpretability, RF and XGBoost add accuracy through ensemble learning, and CNN further advances the deep learning approach for sophisticated pattern recognition. This would create a strong, interpretable, high-performing predictive framework for student performance assessment.

3.5 Explainability with XAI

Thus, as it is predicted now, so high, the chance for Explainable AI (XAI) in education. True, by themselves, these models allow very high performance prediction but to users of these systems especially educators, they work as black boxes denying the user any view on what informs any decision. Predictive modeling spotlights a well-assured source of interpretation, transparency, and trustworthiness in prognosticating student performance in AI, thus empowering educators to make informed interventions through data.

Interpretation of Model Decisions

- Introduction to Feature Importance in brief.
 - As mentioned, Random Forest (RF) or XGBoost models like have an inherent feature importance score to display to users the influential features on student performance. This once again enables the user to establish the contribution of the attendance, assignment scores, peer-interactions among other variables applied in prediction.
- SHAP
 - SHAP provides an explanation on a specific prediction on how much contribution each struck feature had on the result stated. It is general (policy making architectural design) and narrow (individual student cases) interpretations of policies-making AI-driven perception that makes it more practical.
- LIME
 - LIME gives almost explanations through the use of varying input data and tracing the alterations that occur in the predictions. It facilitates more elaborate

models like CNNs and XGboost making them even easier to interpret.

➤ Visualization Techniques

- Examples of other visualization methods to understand the behavior of models: how various factors correlate with one another to affect the performance of the model in predicting students include heatmaps, decision trees or partial dependence plots (PDPS). These methods allow the layperson in the non-technical world to see results.

In this vein, an incorporation of XAI transforms the predictive model further into a more plausible, responsible, and useful tool, thus rendering an efficient utilization of the AI credibility to give fruitful bias reduction and proactive measures to enhance the performance of students. And on its holistic end, XAI eventually seals that gap between the incomprehensibility of AI and common sense to provide human technical education prescience without endangering fairness and transparency in its application.

3.6 Evaluation Metrics

Machine-learning and deep-learning evaluation should be done with the utmost sincerity and precision in order to gain trust in the prediction of student performance. Different metrics evaluate the performance of the algorithms in classifying students into success categories. The ones most relevant to this study are accuracy, precision, recall, F1-score, ROC-AUC, and confusion matrix analysis; a constellation of all of these gives an understanding of whether a model predicts correctly concerning students' performances calling for an academic intervention while minimizing the false ones.

Another of the more widely used evaluation metrics of classification problems is accuracy. Its meaning is the degree of correct predictions made by the model from all predictions. This is calculated as the number of correct positive and negative predictions made by the model divided by the total number of predictions made by the model. Although accuracy serves as a general measure for the performance of various models, in some cases it holds no reliable ground, especially when applied to imbalanced datasets. For instance, if 90% of the students in the dataset are good performers, and 10% of the students are in need of some support through interventions, a model that would predict greatness for everyone would be awarded an accuracy of 90% upon its verdict yet fail abysmally in identifying truly needy students.

To complement this limitation, precision is calculated as the correctness of positive predictions. Precision measures how many of the students classified as successful were, in fact, successful. This is important in cases where we want to minimize false positives, since passing an at-risk student off as high-performing could cost them the chance for intervention. However, precision alone is still deficient in that it disregards false negatives-those students who are at-risk but misclassified as successful.

The opposite of precision is recall which complements and counteracts it with its own concentration by relying on the ability to identify students in need. The main essence of its usage will only cheat the proportion of underachievers who are rightly categorized. When the recall is too large, it will almost certainly result in a higher number of students requiring academic assistance being identified, at the cost of more false positives. Precision and recall should be balanced, however, such that, on the one hand, underachieving students are not neglected, and on the other hand, under-performing students are spared unnecessary interference in an academic setting.

F1 score is a harmonic mean of precision and recall, which provides a balanced measure of evaluation when the distribution of labels of the classes is not uniform. It is utilized largely in a scenario where it is desired to reduce the number of false positives and false negatives. The large F1 score means that the model is not only classifying struggling students accurately but a very small number of them are being misclassified. This consideration is important in forecasting the students performances, as each of early intervention measures and proper classification does count. The other notable measure, the ROC-AUC (Receiver Operating Characteristic-Area Under the Curve) determines the discrimination capabilities of a model between classes. AUCs are therefore very useful in determining the rank of students in terms of the probability of passing in a learning institution. In this regard, the utility of the confusion matrix is that it gives additional information that can be used to assess a model through the presentation of the true positive, true negative, false positive, and false negative counts in the model prediction. This then gives an understanding on what errors the model is committing and how to refine its forecasts.

In this way, such a combination of metrics can assure that the chosen machine learning and deep-learning models make accurate, fair, and interpretable predictions; the idea is to build a model that is as statistically good as possible but also educationally interesting, which enables institutions to detect at-risk students and provide them with necessary support in a timely manner.

3.7 Experimental Setup

The physical infrastructure we have is a combination of hardware, software, and machine-learning libraries making an ecosystem of efficient training, testing, and validation of deep-learning models. Similar to a high-performance computer setting which should be used with large datasets and intricate deep-learning frameworks, the experimentation was conducted on a computer with an Intel Core i5 processor and 8 GB memory to accelerate the training of custom CNN-based architecture deep-learning models. Moreover, GPU support introduces one more level of training-time savings and a higher level of computational efficiency.

Windows OS is highly compatible with a number of deep-learning frameworks in terms of software environment. The programming language that is chosen is Python 3.9 due to its immense machine learning and data analysis support. Jupyter Notebook is where the actual project development work is done to provide an interactive working environment also with version control whereby changes are stored in GitHub to ensure reproducibility.

The libraries used for developing machine learning and deep-learning applications vary widely. Logistic Regression (LR), Random Forest (RF), Linear Discriminant Analysis (LDA), and XGBoost are among algorithms available in scikit-learn. CNN training is done with TensorFlow and Keras, whereas analytical work on social networks (SNA) is carried out using NetworkX. Both SHAP and LIME play a role with respect to explainable artificial intelligence (XAI), thereby ensuring interpretability.

Using optimal hardware-software combination, hence the support framework, targets maximum performance and scalability and implies efficient model training and validation in the context of predicting student performance.

Chapter 4 - Implementation

4.1 Dataset Description

OULAD, or the Open University Learning Analytics Dataset, is the general public access dataset housed at the Open University, UK, and intended to facilitate learning analytics research. This dataset fully presents about 32,593 students who joined seven chosen courses offered in the years 2013 and 2014 so that variables related to success, retention, and engagement can be discovered. Numerous tables include individual student information on demographics, performance in modules, and diverse behaviors in learning from the course. Key context populating demographics investigates the course and between educational outcomes. Course modules hold information on how each course is assessed and structured. In addition to activities undertaken by students, the dataset logs tracks student activities performed within a Virtual Learning Environment (VLE) that would include engagement online with quizzes, forums, and learning materials. Assessments include submissions and scores of students involved in particular assignments and contribute to their overall academic performance.

OULAD has an excellent feature of being prediction- and analytics-ready for timely interventions to improve retention and performance of students. Structured in this way and content-rich, it provides the base for data mining, machine learning, and statistical analysis, allowing educational institutions to improve their learning experience.

4.2 Feature Description

The training period utilized information up to October 2023. Student demographic data structures alongside course structures and activity logs alongside assessment information comprise the main classifications of features found within the Open University Learning Analytics dataset (OULAD) as described below:

- Student Demographics
 - The database assigns individual number codes known as identifiers to unite information across multiple tables.
 - Gender represents a categorical variable that indicates the gender identification of students between male and female options.

- The classification system includes age bands structured to define student populations as "0-35" or "35+". The data helps identify student population distributions according to their age group.
- The data group includes geographical areas called regions which the university has designated for its students across UK territories.
- Student disability self-declaration is documented in the database to perform comparisons among students with disabilities (Yes/No) for learning outcomes and experiences.
- The students displayed their highest previous academic level to the course by showing either their "A Level" or "HE Qualification" degrees.
- Course Modules
 - Code_Module: A unique code applicable to each course module for identification of courses in the dataset.
 - The term code Code_Presentation shows which presentation of a course module exists in a specific academic year and academic semester so different presentations can be distinguished.
 - The length of module presentation refers to the weeks in which the course module is active so students can perform time-based evaluations.
- VLE Interaction Data
 - People can track engagement changes through time by using the time-specific keywords that represent certain days within course module timespans.
 - Site_Type represents the VLE activity type or page type such as forums quizzes resources for interaction classification.
 - The Sum_Click measure shows the complete number of VLE page clicks regardless of user interaction types across every online material or activity.
- Assessment Data
 - The assessment system uses ID_Assessment as an individual reference point for all assessment tasks through their connection to module assignments.
 - The database includes an assessment type field that designates assessment formats through references such as "CMA" for Computer-Marked Assignment and "TMA" for Tutor-Marked Assignment.
 - The date Submitted by students serves as a record which delivers information about how long they spent on their assessments.
 - The assessment score represents an essential metric of student academic

achievement which appears in each assessment record.

➤ Student Assessment Data

- ID_Assessment: Links to the main assessment table for detailed assessment-related data.
- This field records the specific date when students submit their assignments to demonstrate appropriate timing regarding deadlines.
- The assessment scoring system represents the percentage score which aids students' performance evaluation.

➤ Student Registration Data

- The date a student enrolled in a module appears as Date_Registration and reveals student enrollment patterns.
- Date_Unregistration reports both student withdrawal events along with their exact dates thus supporting analysis of student retention rates.
- ID_Student: Links to individual student information.
- Code_Module and Code_Presentation: Links to specific course modules and presentations to trace student interaction patterns

	code_module	code_presentation	id_student	gender	region	highest_education	imd_band	age_band	num_of_prev_attempts	studied_credits	disability	final_result
0	AAA	2013J	11391	M	East Anglian Region	HE Qualification	90-100%	55<=	0	240	N	Pass
1	AAA	2013J	28400	F	Scotland	HE Qualification	20-30%	35-55	0	60	N	Pass
2	AAA	2013J	30268	F	North Western Region	A Level or Equivalent	30-40%	35-55	0	60	Y	Withdrawn
3	AAA	2013J	31604	F	South East Region	A Level or Equivalent	50-60%	35-55	0	60	N	Pass
4	AAA	2013J	32885	F	West Midlands	Lower Than A Level	50-60%	0-35	0	60	N	Pass

Figure 2: First 5 rows of the data

4.3. Exploratory Data Analysis (EDA)

EDA examines determinants in the different tables within the Open University Learning Analytics dataset to have trends, patterns, and insight into how students behave and perform. EDA usually starts with demographics, like age, gender, disability; it may then be possible to understand how big population groups are different in this respect, which may then permit the digging into whether some groups are more active or more successful- are course-completion rates different by age groups? Or do students with disabilities tend to be relatively less or even more interactive with the VLE?

Next comes the more realistic analysis of course characteristics like module length and

assessment that can assist in identifying the probable effect of courses on the student engagement patterns. The generated VLE logs record the interactions of the students and lets the students study the engagement over time and of highest usage times which correspond to the assignments due dates or exams.

Marks and submission dates act as a check to measure the performance of the students with time and how they meet the deadline. The VLE utilization can be compared to the assessment scores; this will give a hint on whether an increased use of VLE would result in higher assessment scores or vice versa. Essentially, Exploratory Data Analysis on this data will be essential to isolate the determinants of paramount interest about academic success, retention and intervention design.

4.3.1 Information About the Data

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 32593 entries, 0 to 32592
Data columns (total 12 columns):
#   Column                Non-Null Count  Dtype
---  -
0   code_module            32593 non-null  object
1   code_presentation      32593 non-null  object
2   id_student             32593 non-null  int64
3   gender                 32593 non-null  object
4   region                 32593 non-null  object
5   highest_education      32593 non-null  object
6   imd_band               31482 non-null  object
7   age_band               32593 non-null  object
8   num_of_prev_attempts  32593 non-null  int64
9   studied_credits        32593 non-null  int64
10  disability              32593 non-null  object
11  final_result           32593 non-null  object
dtypes: int64(3), object(9)
memory usage: 3.0+ MB
```

Figure 3: Information regarding the data

The student information dataset that is an entity of the Open University Learning Analytics Dataset is intended to reveal significant facts concerning each student in regards to the demography, education, and academic outcomes. Included in the traits it has are the gender of the student with place of residence, age band, the highest level of education achieved and disability status. Along with these features are previous study attempts, credits studied, and final results. The dataset includes 32,593 students in various courses and semesters and gives

an idea of student demographic and learning path diversity.

Based on the first glances, one may state that the interesting factors should be: say, `imd_band` referring to socioeconomic status and some of the missing values meant that some students were not in a position to answer this question. It also produces ranges of age in such a manner that the results of academics can be compared to the age. Number of previous attempts is the amount of attempts that the student had attempted in the specific course and this may be a determining force to the final achievement.

The most significant column in this dataset is the final result column where the results of the student are recorded as Pass, Fail, Withdrawn or Distinction. This attribute of outcome is useful because it can then be correlated with other attributes to give information about which aspects contribute to an individual succeeding or failing at school. Hence, this allows discussing potential performance disparities because of such demographics as age, disability, or level of education and understanding the creation of superior student support services.

4.3.2 Statistical Description of Numerical Data

	id_student	num_of_prev_attempts	studied_credits
count	3.259300e+04	32593.000000	32593.000000
mean	7.066877e+05	0.163225	79.758691
std	5.491673e+05	0.479758	41.071900
min	3.733000e+03	0.000000	30.000000
25%	5.085730e+05	0.000000	60.000000
50%	5.903100e+05	0.000000	60.000000
75%	6.444530e+05	0.000000	120.000000
max	2.716795e+06	6.000000	655.000000

Figure 4: Statistical description

The summary statistics of the student information data provide some valuable information of individual study habits and course participation. Student ID is also averaged fairly high because there is a unique identifier system, rather than a trend of a particular group, however, with respect to the feature "`num_of_prev_attempts`" most students-who are typically taking the course the first time with a very low median of 0.16 and a standard deviation of 0.48: median and 75th of 0. The number of students who had the course multiple times, though not more than six times, is limited, and it does not mean that every single student had a good

experience, not to mention that such children might require to be supported even more.

Considering the variable `studied_credits`, the average credits that students have undertaken are approximately 80 credits with a fairly wide range already demonstrated with a standard deviation of approximately 41 credits. The median value of credits at 60 shows that most of the students are part time students. The 75th percentile indicates 120 credits that leave 25 percent of the students in a full load or more intense studies. Most intriguingly, credits of study go up to a total of 655 in maximum which could be attributed to students who take multiple modules or could be simply an outlier in recording of credit. Such statistics would be a basis of analysis in terms as well as of bridging the gap between such analysis and attempts and study load and end results and student retention.

These statistical summaries of the student information data display some rather important findings related to the personal study habits, and involvement in the course. Mean student ID seems to be rather high because there is a unique identifier system instead of a tendency of particular groups. In the feature - "num of prev attempts" most students (median and 75 th percentile are at 0) are indeed actually doing the course the first time as per a mean of only 0.16 and a standard deviation of 0.48. But there are some more who have had the course more than once-the maximum number so far reached may be reckoned at six-at least, marking out a poor student group, who might even be successful with help. Regarding the number of studied credits, students average out about 80 credits and the range is quite large with a standard deviation of about 41 credits. The median credit value is 60; hence, a good number of students are part-time students. The 3/4 th percentile has 120 credits which implies that a quarter of the students are taking up full course loads or even working harder. Most interestingly, the number of credits studied is incredibly high (655) and this may be attributable to the observation that students do multiple modules or-as a measure of credit-this may be an outlier. This number would certainly represent a point of analysis in terms of also correlating previous efforts and study volume with end findings and student retention.

4.3.3 Gender Distribution

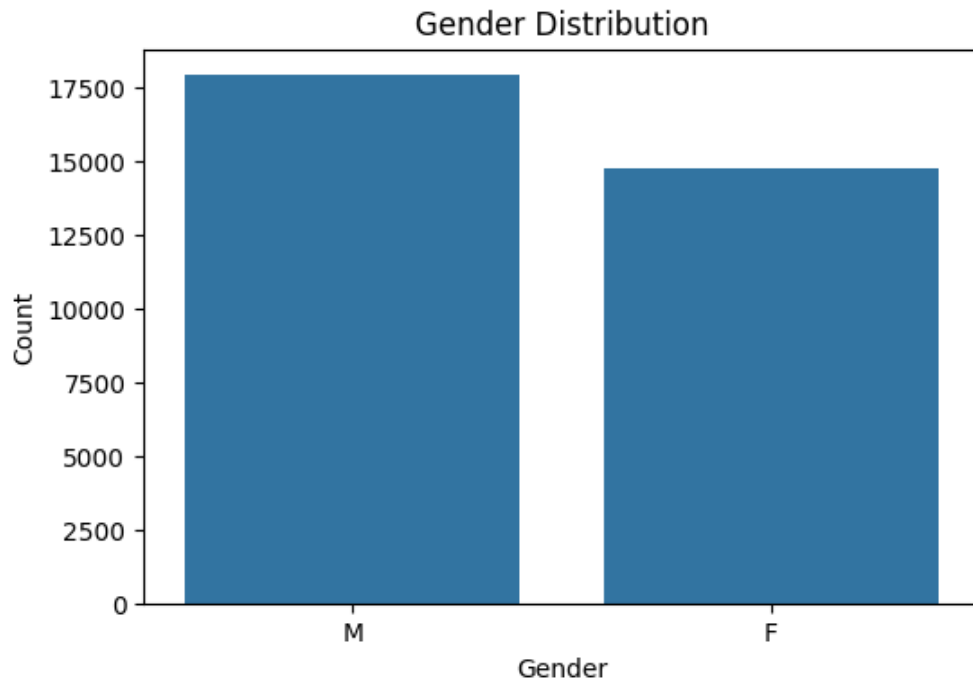


Figure: 5 Gender distribution

The count plot illustrates how the students were distributed by their sex which is somewhat biased towards females since it portrayed the potential reverse of the gendered sample in the uneven sample. This would change likely bias in any observed upon such analyses when explaining any out there and relevant results by gender that might otherwise proceed when it would be necessary that the distribution be understood.

4.3.4 Distribution of Grades

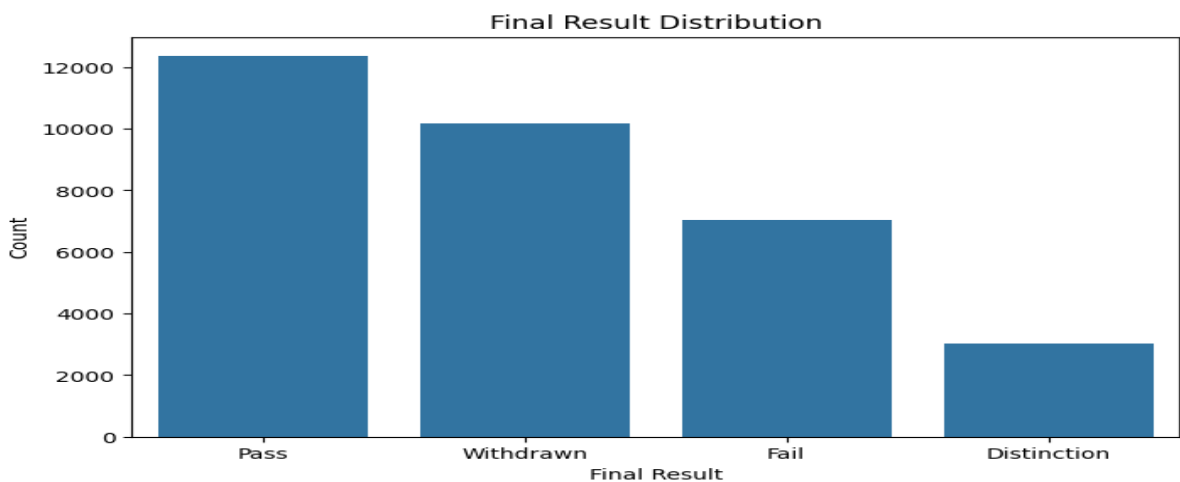


Figure 6: Distribution of grades

The bar chart shows the distribution of the students according to the final results. The

majority of the students were either graded as Pass, then a significant number of students were graded Withdrawn. This high withdrawal rate has the risk of certain challenges that encountered the students in the process of their studies. Fail is not very much, yet it deserves to be mentioned, which means that there is some academic suffering in the student. A small proportion of students have scored at "Distinction" hence supporting the fact that a minority do succeed in their courses. This distribution requires an investigation of the variables that are linked with the withdrawal and failure rate, and potential interventions that aim to improve student performance and reduce the dropout rate.

4.3.5 Missing Value Inspection

This table illustrates the missing values in each of the features in the dataset. There are no missing values in most of the features like code-module, code-presentation, id-student, gender, region, highest-education, age-band, number-of-prev-attempts, studied-credits, and disability and final-result. This means that there is an acceptable level of completeness of the data. The exception is only so id band, which reflects the socio-economic status but 1,111 entries are missing. The imd band missing data might affect any assessment of the socio-economic factors as it creates a gap in understanding of the potential consequences of the socio-economic background on academic achievement. Imputation or addressing these missing values would help to enhance the reliability of data in those analyses.

	0
code_module	0
code_presentation	0
id_student	0
gender	0
region	0
highest_education	0
imd_band	1111
age_band	0
num_of_prev_attempts	0
studied_credits	0
disability	0
final_result	0

Figure 7: Missing value inspection

4.3.6 Distribution of Highest Education

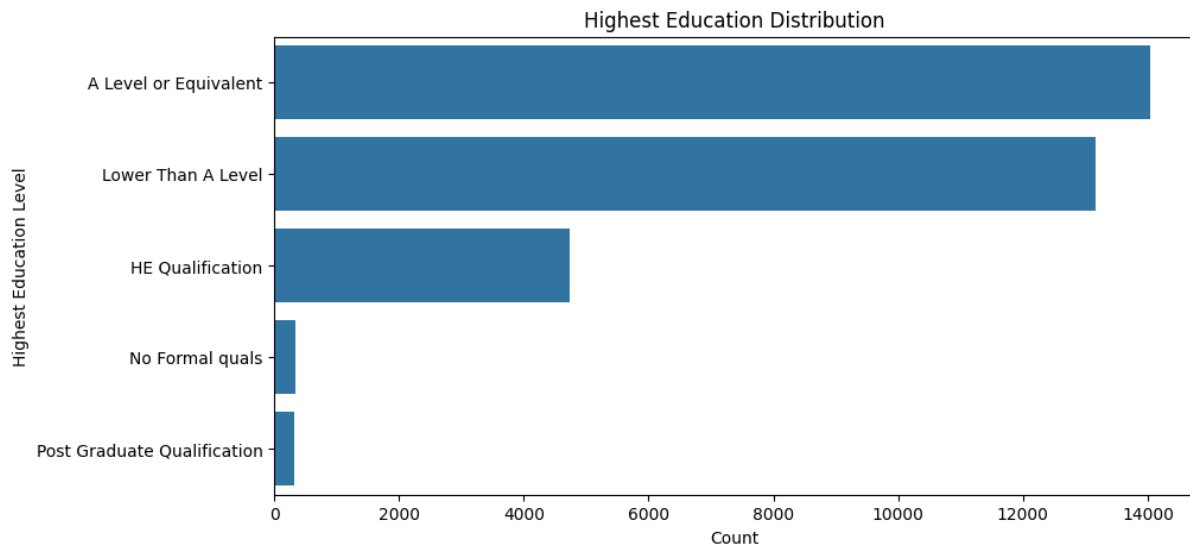


Figure 8: Distribution of highest education

The bar graph presents a clear indication of students' qualification distribution about the highest level of education attained prior to joining. The bulk of students are at a level of qualification roughly akin to A Levels, followed soon after with those below that. HE Qualification (Higher Education) represents a lower percentage of the students whereas the least represented categories are Post Graduate Qualification and No Formal Qualifications. Thus, the evident distribution indicates many students having basic educational backgrounds and fewer attainers of higher academic qualifications. Understanding this distribution is important for analyzing how students relate their past learning experiences with their present performance, thus affecting engagement and retention levels as each of the previous levels has different preparedness and support needs.

4.3.7 Final Result Vs Gender Comparison

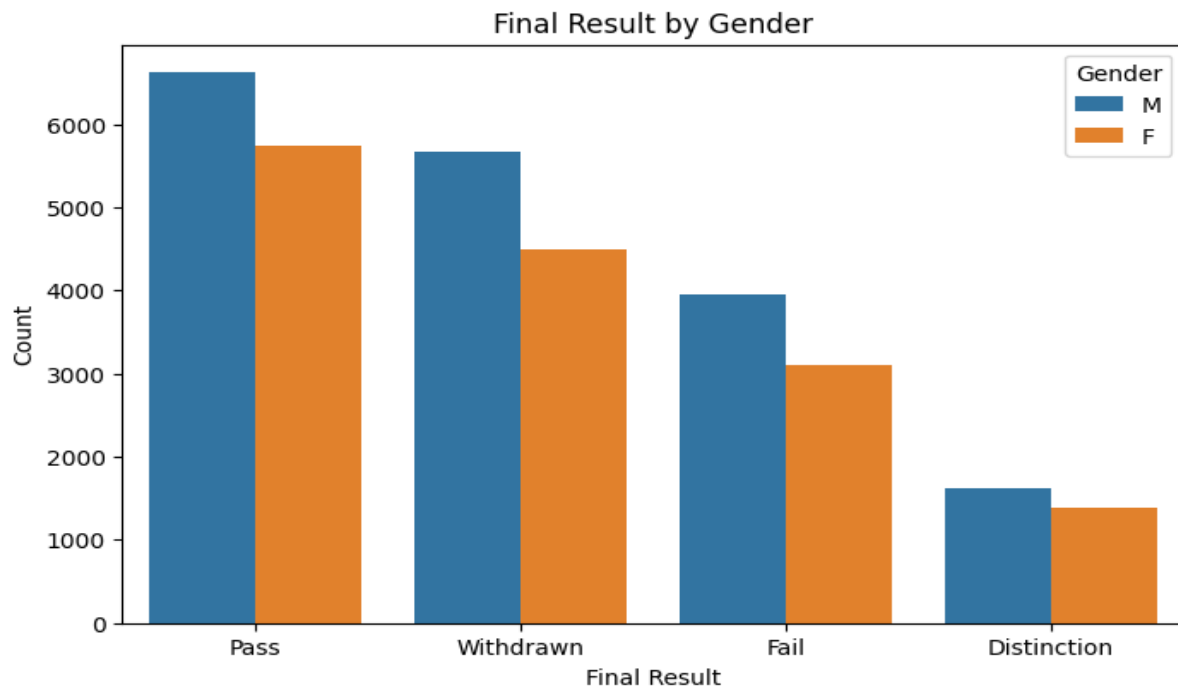


Figure 9: Final result Vs gender comparison

The figure illustrates that final results between males and females show comparable patterns. The total number of students who achieved "Pass" results is high for both gender groups and distribution frequencies of "Fail," "Withdrawn" and "Distinction" match between genders. The analysis shows that gender provides little influence on students' academic success since their performance results align identically between the sexes. The student enrollment distribution of females shows some impact on overall patterns even though it does not influence the success rates between different outcome groups.

4.3.8 Final Result Vs Region Comparison

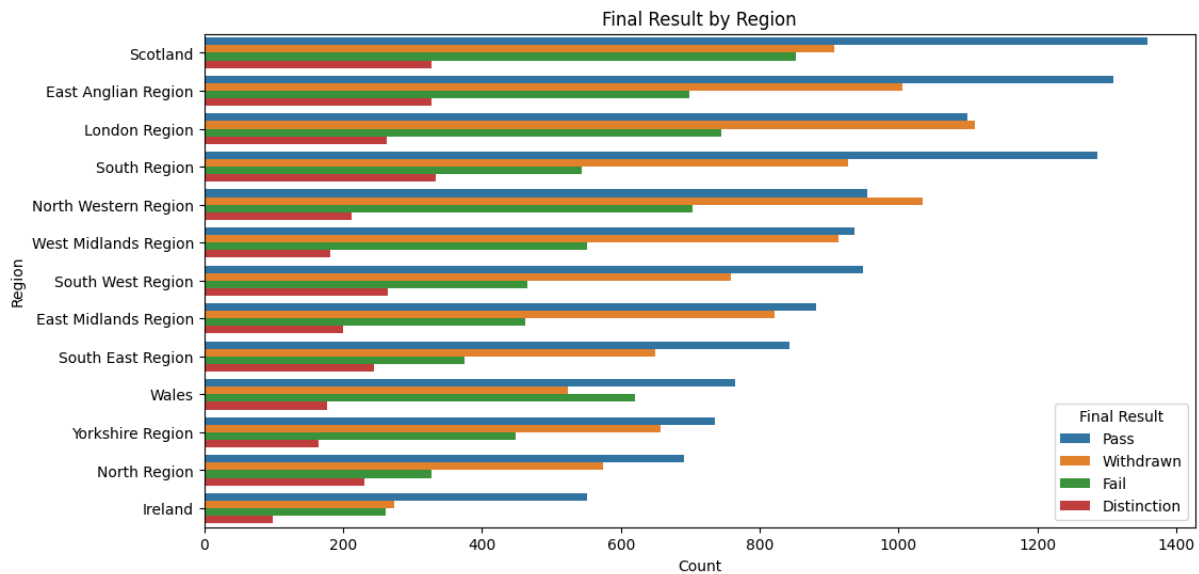


Figure 10: Final Result Vs Region Comparison plot

The graph shows how the final results vary between regions, with differences on the outcome types pass, withdraw, fail, and distinction. Most regions have a high count of pass results and show a majority of equal performances across Scotland, the South, and the North Western regions. However, again there are a large number of similar results from "Withdrawn" especially in East Anglia and the South East, which suggests that there are a large number of students who have dropped out of university in these areas. Very few found a distinction across all regions: thus, it's a minority that usually fills the high achievement. This analysis in the region can allow highlighting differences in student outcomes that may be geographical or may be generated by regional factors affecting student engagement and success.

4.3.9 Pair Plot



Figure 11: Higher Education Aspirants Achieve Better Grades

The pair plot shows two noticeable outliers: One has an unusually high average click count, suggesting extreme interaction with the VLE as compared with other students, while the other shows a student with an exceptional and high number of previous course attempts, in contrast to the other students. These outliers may bias the data, which in turn might lead to biased interpretations. For the sake of consistency of the data and to allow more accurate analysis, these two cases of outliers will thus be discarded. This would maintain a focus on accepted behaviors of students, increasing robustness of findings and interpretations.

4.3.10 Correlation

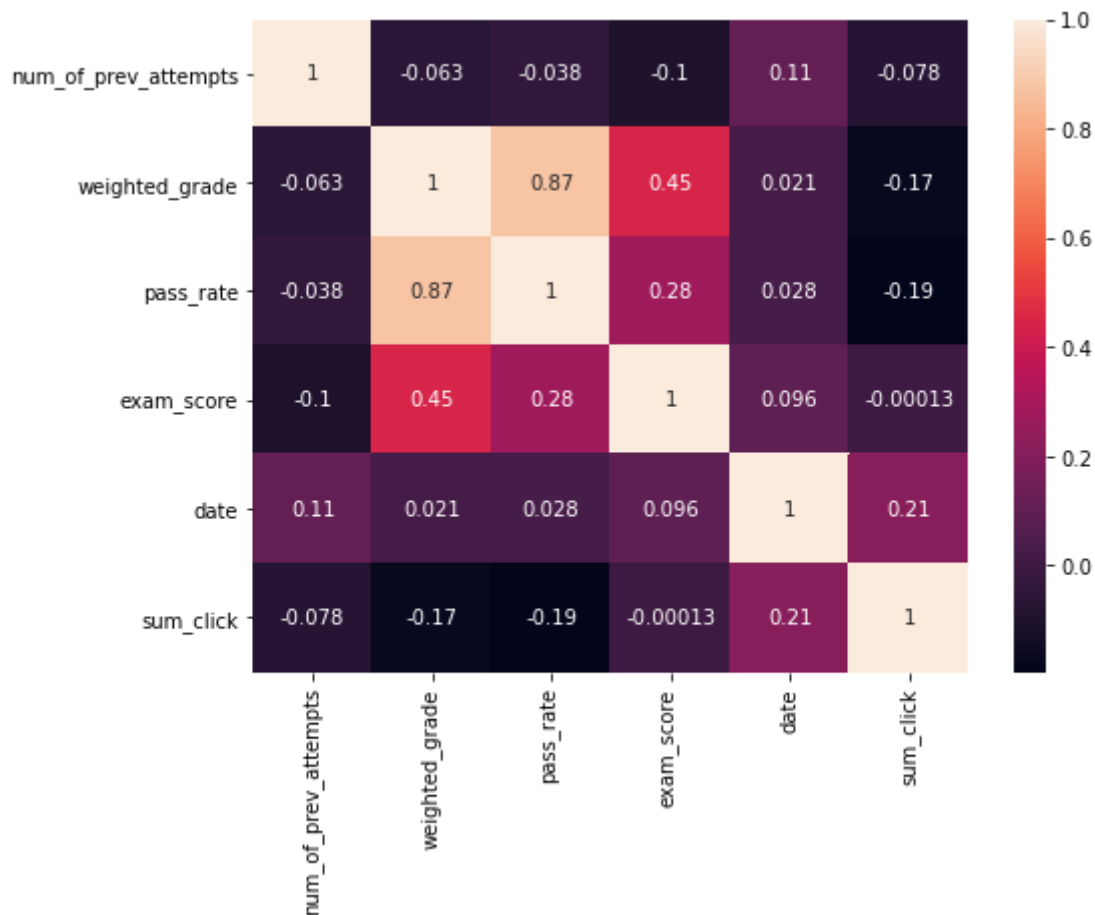


Figure 12: Correlation plot

A heat map visualizes feature correlations so the darkest parts show weak relationships but lighter areas display strong relations. A pair of particular variables demonstrates an extremely strong positive correlation that suggests such variables either link closely or impact one another directly (0.87). The two features demonstrate a moderate strength relationship at level 0.45 which shows weak dependency between them. Most other pairs of features reveal weak linkages or demonstrate no correlation at all in the analysis. The analytical assessment shows which variables strongly influence each other thus assisting the feature and dependency selection process.

4.4 Logistic Regression

What is the progress of the training data until October 2023? This is what we do at the

start of model training with logistic regression; an important algorithm in binary classifications- to predict the performance of a student. First, we used final_result, as a feature combining both weighted_grade and pass_rate, to estimate outcomes for the students overall. We then use the weighted_grade alone as the independent variable to show the effect of the weighted grades on performance. Finally, using the pass_rate alone helps achieve an understanding of its predictive influence on student results. With the aforementioned sequential order of making models, one can observe how each variable impacts the model's predictive power for performance aforementioned.

A system that predicts the chances of a given input belonging to a category is called logistic regression. It determines whether a student is likely to pass or fail based on final_result, weighted_grade, and pass_rate. Logistic regression applies the sigmoid function for modelling the probability output between 0 and 1. The raw scores are transformed into probability by the sigmoid function with nearer to 1 meaning a high possibility of passing and nearer to zero indicating a high possibility of failing.

Such an optimization algorithm finds optimization of model weights by using maximum likelihood estimation-so that the weights of the input features are fit in a way that the difference between the predicted and actual results is minimized. The training of model in logistic regression is done by updating the weights repeatedly until the model fits optimally. By so, such probability analyses establish a limit (typically 0.5) on the basis of which to declare pass or fail whatever aspect of the student population the application is attempting to address. Simply put, it tells you how each feature such as weighted_grade or pass_rate affects performance. Therefore, there is a lot to gain from it as a tool for predicting success in school using this data.

Training Scenario	Precision (Distinction)	Recall (Distinction)	F1-Score (Distinction)	Precision (Fail)	Recall (Fail)	F1-Score (Fail)	Precision (Pass)	Recall (Pass)	F1-Score (Pass)	Accuracy	Macro Avg Precision	Macro Avg Recall	Macro Avg F1-Score	Weighted Avg Precision	Weighted Avg Recall	Weighted Avg F1-Score
Using Final Result	0.84	0.77	0.8	0.86	0.68	0.76	0.89	0.94	0.92	0.88	0.86	0.8	0.83	0.88	0.88	0.88
Dropping Weighted Grade	0.83	0.75	0.79	0.86	0.67	0.76	0.88	0.94	0.91	0.87	0.86	0.79	0.82	0.87	0.87	0.87
Dropping Pass Rate	0.83	0.76	0.79	0.85	0.68	0.76	0.89	0.94	0.91	0.87	0.86	0.79	0.82	0.87	0.87	0.87

Figure 13: Table differentiating the precision, F1-Score, Accuracy

The prediction model achieves satisfactory results in student outcome forecasting which leads to positive outcomes for all cases. The model achieved 88% accuracy together with a 0.83 F1-score (macro average) when evaluating all final_result data elements

(weighted_grade and pass_rate). The model demonstrates excellent predicting power for students who will pass courses based on its high precision and recall rates for the "Pass" category.

When weighted_grade is omitted from the analysis the model accuracy drops slightly to 87% while all F1-scores observe minor decreases. The model maintains a sufficient level of accuracy since weighted_grade improves prediction accuracy although it does not impact its fundamental performance.

By taking out pass_rate the model keeps an accuracy level of 87% that matches near identically the scenario without weighted_grade. The model shows minor performance enhancements when weighted_grade and pass_rate join existing features for measurement. When final_result stands alone it demonstrates strong student outcome prediction ability whether or not weight_grade or pass_rate are left out. The retention of variables leads to greater precision and recall performance for student outcomes that receive Pass or Distinction grades.

4.4.1 Explainability AI for Logistic Regression

SHAP

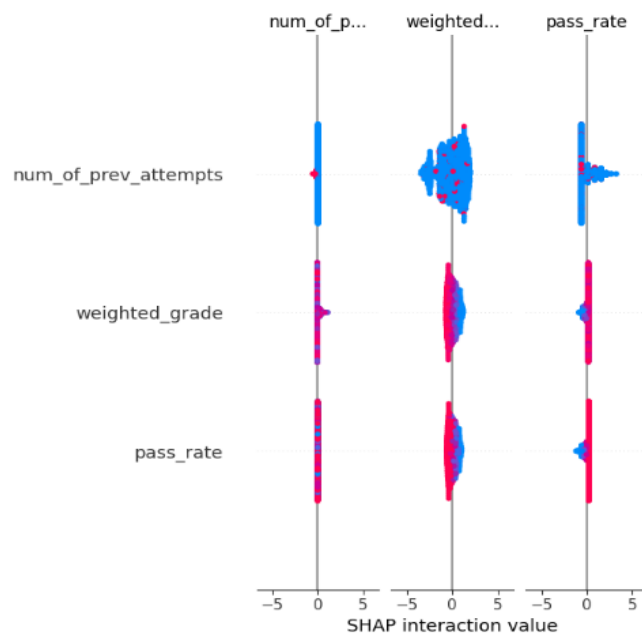


Figure: 13 SHAP for logistic regression

The SHAP (SHapley Additive exPlanation) graph is the visualization tool that explains the output of a machine learning model. It directly helps in understanding the contribution of feature in the prediction made by the model. Here is what you mentioned, in the context of a SHAP graph:

- num_of_prev_attempts: This feature represents the number of previous attempts made by the student. Thus, in the SHAP graph, this feature also explains how it affects the prediction of the model. For instance, it is modeled that more attempts might be negatively correlated with success concerning predicted outcomes.
- weighted_grade: A composite grade could represent various factors in its calculations, and thus, this feature's SHAP value will indicate how changes in weighted grades would impact the model's predictions, whereby higher weighted grades might positively impact the prediction.
- pass_rate: This is likely to be absorbed or represent the former pass rate achieved by students. So, in the SHAP graph, this will also demonstrate how such rates affect model predictions. The higher pass rate is obviously linked to a greater probability of success in the output from the model.
- The SHAP interaction value: The interaction effect among features is shown by this value. It helps understanding how each pair of features actually combines in making predictions (for example, num_of_prev_attempts and weighted_grade) which may not be visible otherwise if each feature is considered alone.

In short, the SHAP graph shows in detail the contribution of each feature and its interaction to the predicted output of the model, making the interpretation and trustworthiness of model decisions easier.

LIME

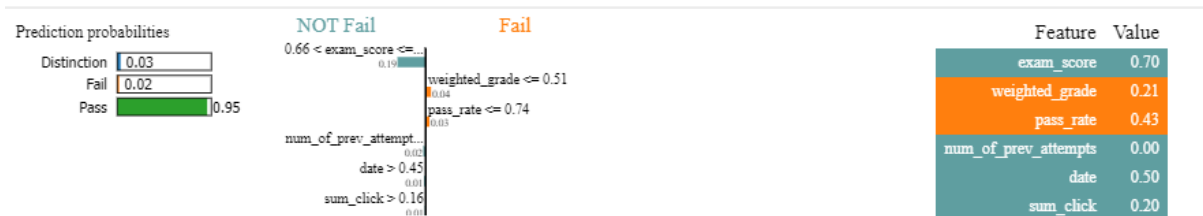


Figure 15: LIME for logistic regression

The image indicates how the prediction probabilities and feature values work together for a classification model predicting student outcome, which may be pass, fail, or distinction. The following elucidates the insights:

- Prediction Probabilities:
 - Distinction: 0.03 (3% probability)
 - Fail: 0.02 (2% probability)
 - Pass: 0.95 (95% probability)

The model predicts that there is a high probability of passing (95%), while the probabilities of the other two outcomes are very low: distinction (3%) and fail (2%).

- NOT Fail Conditions:
 - exam_score: Between 0.66 and 0.04
 - weighted_grade: Less than or equal to 0.51
 - pass_rate: Less than or equal to 0.74
 - num_of_prev_attempts: Greater than 0.45
 - sum_click: Greater than 0.16
 - num_click: Equal to 0.14

These are the conditions that describe feature values associated with not failing (high exam score and weighted grade, with a certain number of previous attempts and clicks, promote not failing).

- Fail Conditions:

- exam_score: 0.70
- weighted_grade: 0.21
- pass_rate: 0.43
- num_of_prev_attempts: 0.00
- sum_click: 0.50
- num_click: 0.20

These feature values are typical for failing, like a low weighted grade and pass rate and no previous attempts.

In conclusion, the image gives a vivid indication of what the model predicts and the feature values influencing the predictions. It indicates the condition under which a student is likely to pass or fail and what are the differing factors responsible for these outcomes.

4.5 Random Forest Classifier

The Random forest is an algorithm used for student performance prediction based on the features of final_result , weighted_grade and pass_rate using Random Forest. Random forest is an ensemble learning method which makes decision trees based on the votes of many trees. These trees are built on random subsets of data and random subsets of features effectively in order to reduce the risk of overfitting and have increasing build power. It aggregates all outputs of decision trees and in case of doing prediction with classification, such as whether a student passes or fails, the final output determines the class with majority number of votes.

Random forest is good for this particular data set evaluation because it is with high-dimensional data, it returns feature importance scores, and above all has resistance against the outliers. All of the above mentioned states also matches to the use of this algorithm in decision-making with respect to the contributions of different actions such as weighted_grade and pass_rate into student outcomes. One major highlight with Random Forest modeling is that it is based on complex interaction among features that offer significant improvement in predictive performance as compared to simple linear modeling.

Training Random Forest using the `final\_result` feature which utilizes both `weighted\_grade` and `pass\_rate` predicts the outcomes with fairly higher accuracy. Further training with either `weighted\_grade` or `pass\_rate` measures how important the individual features might be.

In fact, on the one hand, Random Forest is the model predicting the student's

performance effectively, while at the other end, it provides interpretation on which features were most influential. This can, therefore, enable a teacher to identify the most essential factors contributing to student success and provide a foundation for fine-tuning interventions to assist students.

Training Scenario	Precision (Distinction)	Recall (Distinction)	F1-Score (Distinction)	Precision (Fail)	Recall (Fail)	F1-Score (Fail)	Precision (Pass)	Recall (Pass)	F1-Score (Pass)	Accuracy	Macro Avg Precision	Macro Avg Recall	Macro Avg F1-Score	Weighted Avg Precision	Weighted Avg Recall	Weighted Avg F1-Score
Using Final Result	0.86	0.85	0.85	0.88	0.77	0.82	0.92	0.95	0.93	0.91	0.89	0.85	0.87	0.9	0.91	0.9
Dropping Weighted Grade	0.8	0.82	0.81	0.87	0.76	0.81	0.91	0.93	0.92	0.89	0.86	0.84	0.85	0.89	0.89	0.89
Dropping Pass Rate	0.86	0.85	0.85	0.87	0.76	0.81	0.92	0.94	0.93	0.9	0.88	0.85	0.86	0.9	0.9	0.9

Figure 16: Table differentiating the Training scenario

The Random Forest model can predict student outcomes quite favorably, achieving high accuracies across the three training contrasts. From using the overall final_result data (comprising both weighted_grade and pass_rate), the model manages a maximum accuracy of 91% and a macro-average F1-score of 0.87: it insinuates that a majority of the available features will stimulate the model's discrimination ability among "Distinguished", failed, and passed categories, especially on a rather high precision and recall on the "Pass" category.

When the model is fed without this weighted_grade feature, a very minor percentage of decline occurs, whereby it will now score an accuracy of 89%. Declines on F1-scores can be expected on all categories too. This tells us that the weighted_grade feature is an effective predictor for such a model-a positive contribution towards a better performance but not of utmost consideration as it is possible to keep the accuracy high without it.

A further drop of pass_rate has implied a slight accuracy drop up to 90%, which also posits a significant contribution by pass_rate to the predicted power of the model. As with weighted_grade, the performance isn't compromised critically; it will still continue to do good classification jobs even without it.

As such, random forest models are very good and powerful for this job and internally provide information about which of the features impacts student performance predictions most. The model can depend on predictions without them, thus keeping it as a reliable tool to predict outcomes and to help educators by identifying students who may need extra support.

4.5.1 Explainability AI for Random Forest

SHAP

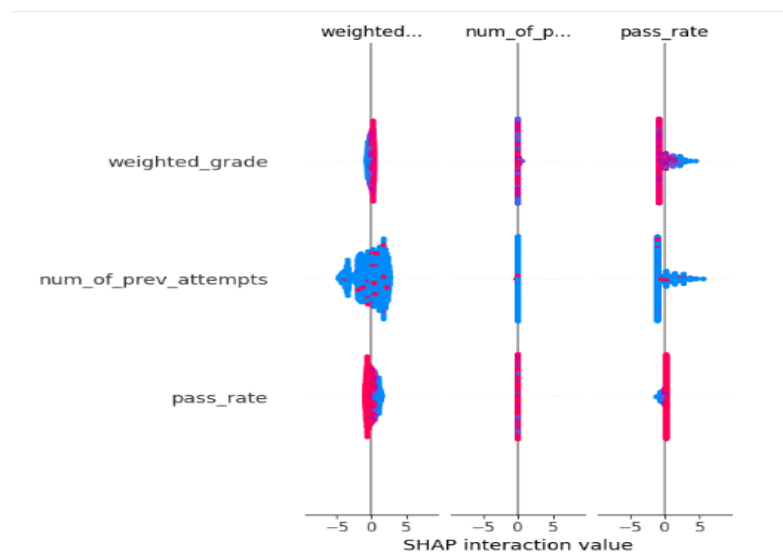


FIGURE 14: SHAP FOR RANDOM FOREST

This seems to be a SHAP interaction value image related to a Random Forest model. SHAP values are used to explain the output of any machine learning model; they indicate how each feature believes in contributing towards a particular prediction. Below are the definitions of components accordingly for a random forest model:

➤ Features:

- **weighted_grade:** This most likely is a composite score or grade that has been weighted according to certain factors. This shows through this SHAP plot how changes made on this weighted grade affect the model predictions.
- **num_of_prev_attempts:** Number of attempts a student has made before. SHAP now gives such value where the number of such attempts might negatively impact the prediction it makes.
- **pass_rate:** Pass rate historically. This value indicates how this in turn plays into the model's calculation, with the higher it is actually resulting in more favorable predictions from it.

➤ SHAP Interaction Values:

- SHAP interaction values show how the interaction between pairs of features affects the model predictions. For example, the **weight_grade** and **num_of_prev_attempts** might reveal that students with higher grades and fewer attempts are more likely to pass.

➤ SHAP Value Range (-5 to 5):

- The x-axis ranges from -5 to 5, which means the effect of each particular feature is going to have on the model output. Positive SHAP values show that the feature increases the prediction score (likelihood of passing), whereas negative ones indicate a feature decreasing the score of prediction.

➤ Random Forest Context:

- SHAP values then help to individualize the contribution of each feature in the tree ensemble. He gets a lot of predictions short from a lot of trees. Hence SHAP values easily measure feature importance and interaction effects at the level of the prediction process.

In short, the picture becomes very informative about how features and interactions between them contribute to the prediction of a Random Forest model. Qualitatively, it lets one see which features are more important and how they interact to affect the result-the model outputs are now interpretable in secured form.

LIME



Figure 15 : LIME for random forest

LIME (Local Interpretable Model Agnostic Explanations) was used to obtain both prediction probabilities and feature values for the Random Forest model. It is a predictive explanation technique for machine-learning models that approximates the model locally by an interpretable one. It tells the overview in summary about the insights:

➤ Prediction Probabilities:

- 0: [0.00] (0%)
- 1: [0.00] (0%)
- NOT 1: 0.99 (99%)

- The conclusion reached by the model is that there is a very high probability (99%) associated with the outcome being "NOT 1," while there are marginal probabilities for the other classes.
- Conditions for NOT 1:
 - exam_score: Between 0.50 and upper limit. 0.14 is attributed to it.
 - weighted_grade: Between 0.69 and upper limit. 0.03 is attributed to it.
 - rate_of_pass: Within 0.74-1.00. It has an attribution value of 0.07.
 - sum_chick: sum_chick less than or equal to 0.07 has an attribution of 0.02.
 - date: Between 0.37-0.45. It has an attribution value of 0.01.

These conditions bear out the characteristic values and forms their contributions to predicting NOT 1. An example is: a high score from exams and weighting increases in both weighted grades and pass rates indicate NOT 1.

Feature Values:

- Centering on the specific values mentioned above, these are the feature values considered for this instance of prediction.
 - exam_score: 0.64
 - weighted_grade: 0.80
 - pass_rate: 1.00
 - sum_chick: 0.05
 - date: 0.44
 - num_of_prev_attempts: 0.00
- LIME Context:
 - LIME explains its prediction model through a local approximation around the instance of interest by identifying which features are the most decision-relevant for that particular prediction, and how those features 'helped' the model come to the outcome predicted; LIME suggests that exam_score, weighted_grade, and pass_rate have significant contributions to the prediction of NOT 1.

It would mean that the said image accurately conveys how each feature affects the Random Forest model prediction for any given instance, providing the necessary interpretability weight for model predictions concurrently identifying key features and values that led to high probability with respect to being NOT 1.

4.6 LDA Classifier

Linear Discriminant Analysis (LDA) is a supervised classification scheme, mainly used to predict an outcome variable with linearly separable data. LDA gives good results if the aim is to maximize the separation between several classes. In contrast to Random Forests that are based on an ensemble of decision trees, LDA presumes that data when regarded separately into classes follow Gaussian distribution within each individual class.

It essentially works on the different linear combinations of input features, which when applied together give maximum separation for actual classes. It provides maximum variance in separating classes subjected to some constraints on the variance within classes with respect to that class. Therefore, LDA is found to be especially useful in the case of those datasets where understanding the relationship between features in less-dimensional space helps in interpretability.

Concerning the prediction and classification of students, LDA can explain how the final_result, weighted_grade, and pass_rate features relate to one another to classify students as passing or failing. The procedure involves calculating the probability that each point belongs to a class based upon the feature distribution and assigning the data point to the class having the highest probability.

The simplicity and computational efficiency of LDA recommend it for a moderate-sized dataset, where relationships between predictors and outcomes are fairly simple. Further, LDA gives interpretable results about factors contributing most to class separation. For educators, these insights could assist in identifying key areas that require remedial support for students, thus refining interventions that could enhance academic outcomes on a larger scale.

Metric	Precision (Distinction)	Recall (Distinction)	F1-Score (Distinction)	Support (Distinction)	Precision (Fail)	Recall (Fail)	F1-Score (Fail)	Support (Fail)	Precision (Pass)	Recall (Pass)	F1-Score (Pass)	Support (Pass)	Overall Accuracy	Macro Avg	Weighted Avg
With final_result	0.81	0.87	0.84	266	0.74	0.78	0.76	198	0.92	0.89	0.91	1021	0.87	0.82	0.88
Dropping final_result	0.82	0.88	0.85	266	0.75	0.78	0.77	198	0.93	0.9	0.91	1021	0.88	0.83	0.88
Dropping pass_rate	0.8	0.87	0.83	266	0.76	0.8	0.78	198	0.93	0.89	0.91	1021	0.88	0.83	0.88

Figure 19: Table differentiating the metric

Insights about the performance of different configurations of the LDA classifier provide understanding of how the features final_result, pass_rate, and weighted_grade affect the model. The model with final_result performs with an overall accuracy of 87%, with the Pass category

getting high precision and recall at 0.92 and 0.89, resulting in an F1-score of 0.91, which indicates that the model performs fairly well in predicting students that are likely to pass. The Distinction category behaves well with precision at 0.81 and recall at 0.87, scoring an F1 of 0.84. The Fail category, on the other side, performs slightly lower at precision 0.74 and recall 0.78, hence starting to show some difficulty in identifying students at risk of failing.

Removing `final_result` improves the overall accuracy slightly to 88% against increased precision and recall for all categories. With precision at 0.93 and recall at 0.90, Pass performs well, whereas Distinction benefits from increases in precision to 0.82 and recall to 0.88. For Fail, improved precision to 0.75 and recall to 0.78 provide our conclusions with more confidence that removing `final_result` smoothes out the discriminating power of the classification.

Likewise, removing `pass_rate` also has an overall accuracy of 88%. While the Pass category continues to perform strongly, with a precision of 0.93 and recall of 0.89, there are slight declines in the performance of Distinction and Fail categories. Nevertheless, the overall macro and weighted averages remain unaffected across configurations, demonstrating the stability of the LDA classifier with varying feature sets.

4.6.1 Explainability AI for LDA

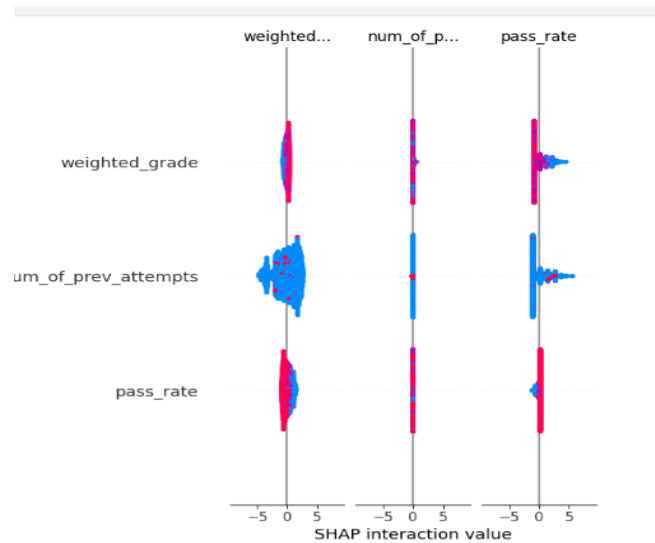


Figure: 20 SHAP for LDA

So, this could mean the SHAP interaction values, but in case of Linear Discriminant Analysis (LDA), they are interpreted somewhat differently because LDA is a classification and dimensionality reduction approach. Still, one could use the SHAP values for extracting relevant information about features' contribution to the model's predictions. Here's what you would understand by components in case of LDA:

➤ Features:

- **weighted_grade:** This is a score or grade represented using certain weighting factors composite; in other words, the SHAP plot shows changes in the weighted grades and how they affect model prediction on such weightings.
- **num_of_prev_attempts:** This feature shows a count of all prior attempts made by the student. The SHAP value for this indicates its influence on the prediction, and more attempts could yield somewhat negative results.
- **pass_rate:** This feature is the historical passing rate, and the SHAP value explains how this affects the model's predictions: a higher pass rate could lead to a prediction being more favorable.

➤ SHAP Interaction Values:

- The SHAP interaction values show how the interaction between pairs of features affects the prediction of the model. For example, the interaction between **weighted_grade** and **num_of_prev_attempts** might show that in general students with a higher grade and a few number of attempts were bound to pass.

➤ SHAP Value Range (-5 to 5):

- And according to that, the x-axis is from -5 to 5 and reflects the contribution of each feature toward the prediction made by the model. That is, a higher positive SHAP value means that the feature adds to the prediction score (e.g., probability of passing), while the latter may be interpreted as reducing the score of prediction.
- LDA is a model that tries to find that linear combination of features that best separates the classes. This is where SHAP values can help in explaining the contribution of each feature for this separation.
- The SHAP interaction value helps in understanding how features interact with one another to affect the classification decision, thus aiding in understanding the decision boundaries that are created under LDA.

In a nutshell, the figure presents information regarding how individual features work either alone or together to produce the predictions of an LDA model. This reveals which features were most influential and how they interact with one another in influencing the outcome in such a way as to create an interpretable decision for the model. This is very helpful in the study of the linear combinations on which LDA relies to separate the classes.

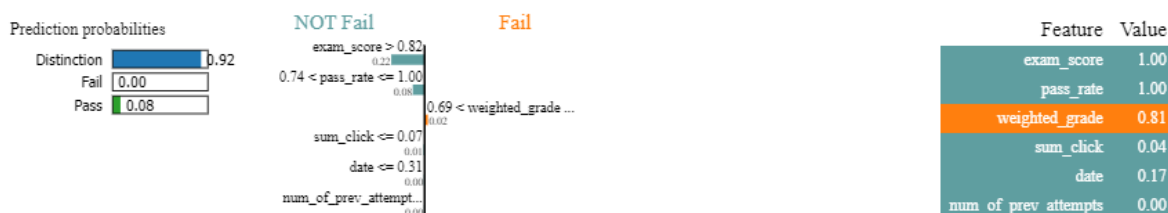


Figure: 21 LIME for LDA

The Prediction Probabilities and Feature Value Representations of a Linear Discriminant Analysis (LDA) Model: Basically, LDA is a statistical analysis classification technique that interpretation the model of predictions by considering contributions of various features as thrown by this figure. Here are all the key takeaways:

- Prediction Probabilities:
 - Distinction: 0.92 (92% probability)
 - Fail: 0.00 (0% probability)
 - Pass: 0.08 (8% probability)

Thus, according to the model, it is a prediction that there can be a high probability of

92% that the outcome is "Distinction" while no probability exists for "Fall" plus an additional very low probability of 8% for "Pass".

➤ Conditions under which somebody would NOT Fall:

- exam_score: Greater than 0.82
- pass_rate: Between 0.74 and 1.00
- weighted_grade: Between 0.69 and a certain upper limit
- sum_click: Less than or equal to 0.07
- date: Less than or equal to 0.51
- num_of_prev_attempts: Specific condition not fully visible

All of these above conditions refer to feature values that tend to be associated with not falling. For instance, a higher exam score in combination with pass rate and weighted grade suggests that one is less likely to fall.

➤ Conditions of Fall:

- weighted_grade: Between 0.09 and a certain upper limit

This condition indicates the feature value that tends to characterize a fall. In other words, low weighted grade suggests falling.

➤ Feature Values:

- exam_score: 1.00
- pass_rate: 1.00
- weighted_grade: 0.81
- sum_click: 0.04
- date: 0.17
- num_of_prev_attempts: 0.00

These are the actual feature values for the instance being predicted. They provide context to the above conditions.

➤ LDA Context:

- LDA is looking for a linear combination of features that best separates the classes. The high probability of "Distinction" is attributed to the high values of exam_score and pass_rate, and also moderate value of weighted grade.

These conditions and feature values reveal how the model differentiates between classes with respect to linear combinations of these features.

4.7 XGBoost

XGBoost is among the most recognized machine learning algorithms that are applied in performing classification and regression tasks because it is fast and accurate. XGBoost (Extreme Gradient Boosting) has been applied in this study to predict student performance using variables like final_result, pass rate and weighted grade. It is a gradient-boosting model that trains an ensemble of decision trees as it enhances itself with each update of specifications of the existing trees. It maximizes a loss such as log loss to perform classification and uses weights so that misclassified samples are more highlighted.

The reason is that the scope of XGBoost is optimized on geared task performance with askew datasets, and can learn complex feature interactions, and avoids overfitting through premature termination and regularization of the model. The algorithm divides data fragments to construct weak learners of the individual outputs assembled into powerful models. Parallelized tree construction and gradient-based optimization enable XGBoost to guard in a new manner to achieve speed up calculation and enhancing the predictive power of the model.

XGBoost certainly delivered high classification results to this project, as well as feature importance scores that needed to identify the mechanism of weighted grade and pass rate in predicting the results as Pass, Fail, or Distinction. Evidently, it contributes to the listing of reasons that justify the use of XGBoost to tackle the multifaceted problem of predicting the performance of students.

Metric	Accuracy	Distinction Precision	Distinction Recall	Distinction F1-Score	Distinction Support	Fail Precision	Fail Recall	Fail F1-Score	Fail Support	Pass Precision	Pass Recall	Pass F1-Score	Pass Support	Macro Avg Precision	Macro Avg Recall	Macro Avg F1-Score	Weighted Avg Precision	Weighted Avg Recall	Weighted Avg F1-Score
With final_result	0.9	0.87	0.8	0.84	266	0.88	0.75	0.81	198	0.9	0.95	0.93	1021	0.89	0.83	0.86	0.9	0.9	0.9
Dropping final_result	0.88	0.82	0.79	0.8	266	0.88	0.76	0.82	198	0.9	0.93	0.92	1021	0.87	0.83	0.84	0.88	0.88	0.88
Dropping pass_rate	0.89	0.87	0.81	0.84	266	0.85	0.74	0.79	198	0.9	0.95	0.92	1021	0.88	0.83	0.85	0.89	0.89	0.89

Figure 22: Table differentiating the Metrics

Multiple XGBoost classifier configurations were assessed for student performance prediction while revealing how feature selection affects both model accuracy and target variable prediction metrics. The accuracy level of Configuration 1 reaches 90%. XGBoost predicts the Pass class excellently through high F1 metrics at 0.93 and precision at 0.90 and recall at 0.95. This demonstrates strong potential to identify those groups successfully. The Distinction class demonstrates performance metrics of 0.87 precision, 0.80 recall that result in an F1-score of 0.84. The statistical indicators for Fail category demonstrate less ideal performance since the precision score stands at 0.88 and the recall score reaches 0.75 while the

F1 metric stands at 0.81. The District shows the most problematic results in this configuration since all metrics including precision, recall, and F1 fall at 0.82, 0.79, and 0.80 respectively. The Pass category together with Fail category demonstrate strong performance through their precision values at 0.90 and 0.88 respectively. The third experimental setup maintains an 89% accuracy rate which distributes performance scores between all categories because the Pass category receives highest marks but the recall measurements for Distinction together with Fail are slightly less effective than the first experimental setup. The XGBoost model proves reliable in various arrangements because it maintains excellent scores on both macro and weighted average values thus establishing its reliability for student performance forecasting.

4.7.1 Explainability AI for XGBoost

SHAP

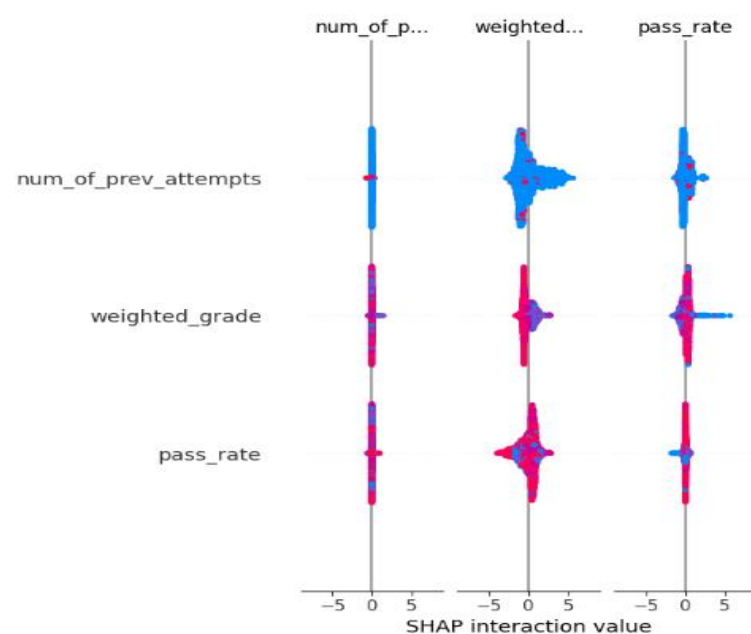


Figure 23: SHAP for xgboost

Typically to be viewed as SHAP interaction values helpful in interpreting outputs from any machine-learning model, particularly those from the XGBoost family of algorithms. In simpler words, we can say that each feature contributed to the prediction in the model using SHAP value.

In this plot:

- num_of_prev_attempts: It describes how many previous attempts had already been made by the student. The SHAP values show how much this feature is pushing the

model's prediction. A positive SHAP value means that higher attempts are leading to an increased predicted outcome, say-increased chances of passing; inversely, it would mean a negative SHAP value.

- **weighted_grade:** This most likely means a weighted score or grade. The rationale is that, as this score increases, so does the prediction made by the model (thus, increasing likelihood of passing).
- **pass_rate:** This most likely refers to the historical pass rate of the course or student. The SHAP values here show the extent to which this feature contributes to the prediction of the model. Higher values on pass rate may affect the prediction in a positive direction.

It is an interaction with SHAP values that indicates plot interactions between many features affecting model prediction. For example, it could tell how the two features `num_of_prev_attempts` and `weighted_grade` work synergistically together on modelling the outcome as compared to the scenario where each one is acting alone on the model.

Thus, the plot gives indication to understanding the great interplay between features in an XGBoost model, and thus brings light along each of their contributions in determining final prediction.

LIME

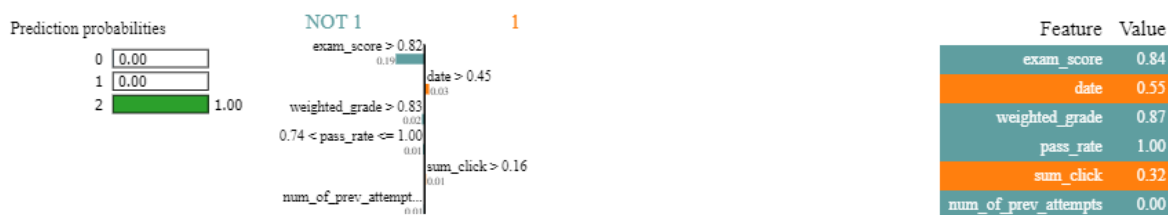


Figure 24: LIME for XGBoost

Your analysis above seems to clearly show how the decision process is made by the ID's the XGBoost model for multi-class classification, understanding from the above interpretation.

- **Prediction Probabilities:** Then the model does a prediction on three possible classes (0, 1, 2) and below is the prediction probability that indicates:
 - Class 0: 0.00 probability (not predicted)

- Class 1: 0.00 probability (not predicted)
 - Class 2: 1.00 probability (predicted with high certainty) The model is confidently predicting Class 2 as the correct class.
- Features and Their Values:
- exam_score (0.84): It seems this feature is contributing considerably to the prediction since it carries a high value of 0.84 that indicates high exam score (>0.82) as the parameter in predicting Class 2 by the model.
 - date (0.55): This feature value is average (0.55); according to the decision tree path, it indicates that when the date feature is higher than 0.45, the model tends to Class 2. It refers to something such as submission date or any temporal factor affecting a prediction.
 - weighted_grade (0.87): This feature weighted grade is very high in strength for contributing to the prediction. As the value is high (0.87), it indicates that probably the student's performance in studies or some other weighted measure counts high in prediction of Class 2.
 - pass_rate: Value 1.00 indicates a clear cut effect of this criterion on the model's decision. If pass_rate is greater than 0.74, the model is leaning with a prediction towards Class 2.
 - sum_click: It indicates moderate value contributing to prediction as 0.32. Although not as much as many features' contributions, it adds a bit to the model decision. Here, the decision threshold above which a shift towards Class 2 happens is above 0.16.
 - num_of_prev_attempts (0.00): The attribute gives a score of 0.00 and shows the attribute not as representative for any decision on the model in this case. If this were somehow related, possibly some other classification would have been held in prediction.
- Decision Path and Impact:
- We have predictions for Class 2 because of a few key elements: exam score being more than 0.82, weighted grade greater than 0.74, and pass rate up above 0.74. The sum clicks here are also contributing, though at a lower threshold, towards Class 2 prediction. The features and their thresholds make the model predict Class 2 with much confidence.

The features exam score, weighted grade, and pass rate were indicative of Class 2 with the greatest effect on the decision.

Summary:

- The model predicts Class 2 with full confidence using primary drivers like exam score, weighted grade, and pass rate.
- The impact of other features like sum clicks and the number of previous attempts may be lower, yet still existent.
- With this decision path, it can be seen how certain thresholds across certain features affect the output, thereby giving an insight into which features contribute majority in the prediction.

4.8 CNN

They are the newest discovery in machine learning research, which gives a definition to that system having capacities such as identification in pattern extraction from complex data sets. In this project, CNNs will analyze student local performance and categorize them into pass and distinction for classification purposes. This process leads to defining the success of students versus the risks of failing students.

CNNs are organized into certain layers, designed to take in the data and extract or learn features that are hierarchical in nature. Convolutional feature extraction identifies certain patterns in terms of `weighted_grade` and in `pass_rate`. Pooling layers enable the shrinkage of the dimensions of a matrix along the spatial axes while holding onto the most important features of the matrix to gain computational efficacy. Fully connected layers will integrate learned features that are trained into accurate classifications.

This mixed-class approach thus enhances the interpretability but also efficacy of the model by putting the outcomes in the language of success and failure. Among the distinctive features of CNNs, there is the fact that they can be used to model extremely complicated relationships, which is why it becomes much easier to model the interaction between final result and pass rate. Due to the high power of CNNs, the predictions will be good even in the presence of noise or less than perfect data.

Therefore, the information obtained on the basis of these assumptions may be used by instructors to learn about the real-life processes that motivate shifts in the achievements of learners. That will in turn result in planning certain changes to get the best out of the at-risk students and strengthen the power of those already on the path of success. The further

scalability and the accuracy of the CNN model enhances its potential usage as a useful educational analytics tool.

Figure 25: Table differentiating the metrics

Metric	With final_result	Dropping final_result	Dropping pass_rate
Precision (Class 0)	0.92	0	0
Recall (Class 0)	0.52	0	0
F1-Score (Class 0)	0.67	0	0
Support (Class 0)	202	202	202
Precision (Class 1)	0.93	0.86	0.86
Recall (Class 1)	0.99	1	1
F1-Score (Class 1)	0.96	0.93	0.93
Support (Class 1)	1283	1283	1283
Overall Accuracy	0.93	0.86	0.86
Macro Avg (F1-Score)	0.81	0.46	0.46
Weighted Avg (F1- Score)	0.92	0.8	0.8

The results of performance measures differed significantly when the classifier was analyzed in three different settings-with final-result, dropping final-result, and dropping pass-rate.

Final result It has demonstrated high overall accuracy (i.e., 93%), which reflects the level of strength of the model to distinguish between the two classes. Class 1 (Pass and Distinction combined) did notably well and had 0.93 precision and 0.99 recall, which gave the F1-score of 0.96. Class 0 (Fail), in contrast, has a much lower recall of 0.52, a number that suggests that it is difficult to identify failing students, but has reasonable precision (0.92).

Dropping final_result: In this case, the accuracy drops to 86, with the performance of class 0 (Fail) dropping to zero in precision, recall and F1-score. This outcome suggests that the

existence of the feature `final_result` is a necessary condition of the model to be in a position of differentiating between failing students. Class 1 retained the full recall (1.00) and also a good F1-score (0.93) meaning that it was a strong predictor of those who were to be termed successful.

Dropping `pass_rate`: The outcome is the same as in the Dropping `final_result` scenario, and again class 0 showed no predictive power and an average value of 86%. This further underscores the significance of the features that have been removed which make the model balance its predictions when there are two classes.

These findings emphasise the importance of proper feature selection in order that model predictions are to balanced and reliable ends.

4.8.1 Explainability AI for CNN

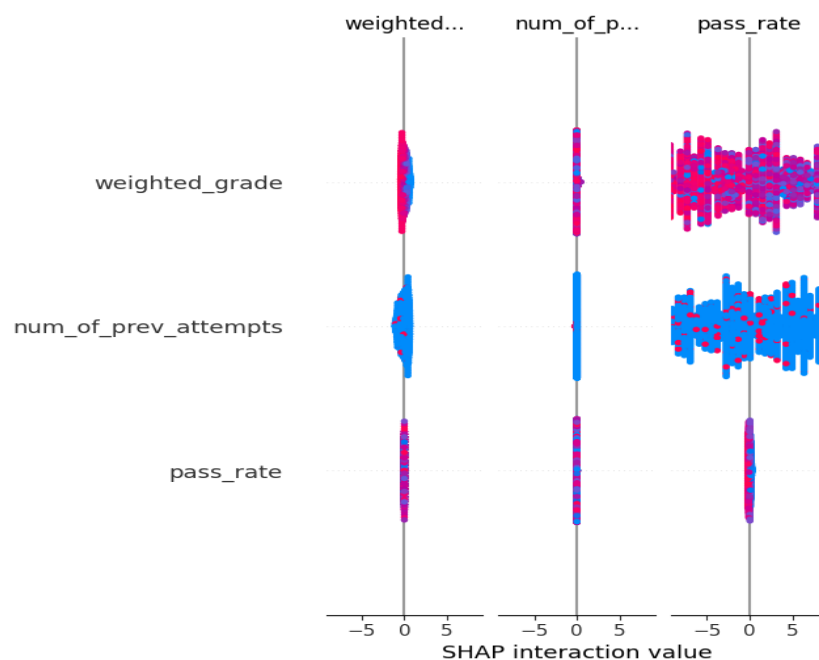


Figure26 : SHAP for CNN

SHAP interaction values are utilized in the interpretation of convolutional neural networks (CNNs) for the contributions that each feature can have on the model's predictions. These can be understood when associated with CNN as follows:

- Features:
 - `weighted_grade`-the feature most probably adjusted grade or score due to some factors. In the case of the CNN model, this use of SHAP values tends to be by assessing how changes in weighted grades affect the prediction by the network.

- num_of_prev_attempts-the total number of previous attempts by a specific student. In this case, the SHAP value shows how this feature affects the prediction, as there is an increasing number of prior attempts holding potentially negative effects on the predicted outcome.
- pass_rate-this feature is historical measuring of the pass rate. So the SHAP value of this variable states how the value influences model decision making, such that a higher pass rate would translate to a more favorable prediction.
- SHAP Interaction Values:
 - SHAP interaction values in CNN explain how those pairs of features interact in affecting the outcome's prediction by the model. For example, the interaction between weighted_grade and num_of_prev_attempts indicates that if a student has a high grade and has attempted fewer times, the outcome is probably a positive prediction.
- SHAP Value Range (-5 to 5):
 - Represented along the x-axis of the plot are the values, which ranged from -5 to 5 and denoted as each of the features-effects (or contributions) to the output given by the model. Thus, a positive value for a feature would imply that such a feature increases the predicted score of the model (meaning higher probability of passing), while negative values indicate a lowering of the predicted score.

This is typically done by convolution on the spatial data, usually images or grid-like data in CNN context. SHAP calculates how individualities (or regions in an image) affect the model prediction, to see which areas in the image or features are counted as most important in its decision.

The result of feature or region combination interaction further clarifies how model input actually derives its output. The inside knowledge can also help in understanding more complex boundaries of decisions that this particular CNN model draws.

4.9 Federated Learning

Federated Learning (FL) is an algorithm of decentralized machine learning in which many clients work to collectively train a shared model without requiring them to share raw data. Rather, each client updates the model on their own data locally, and then only transmits the weights which were updated back to a central server. The server will then aggregate these weights- most commonly with Federated Averaging (FedAvg) and distributes them to all the clients to start the next training round with.

The proposed study entailed training a fully connected neural network in a simulated FL setting aided by five clients. Local training was done by each client after which weight aggregation was done at the server. Eight federated rounds of this cycle took place. The results of the model in these rounds demonstrated that the accuracy varied between 56% and 83.3 %.

Such difference in accuracy would be caused by disparities in distribution of data among clients (non-IID data), scarcity of local training data, and randomness of parameter initialization in each round. However, FL has a major privacy advantage and can permit learning through the use of geographically dispersed sources of information without data sharing.

To have greater accuracy, one can consider some of the techniques, such as high local epochs or learning rate tuning or regularization or more advanced forms of aggregation. Despite the observed fluctuation, the results suggest the potential of FL in the training of models in a privacy-preserving manner in the setting in which the input data are challenging to be centrally located.

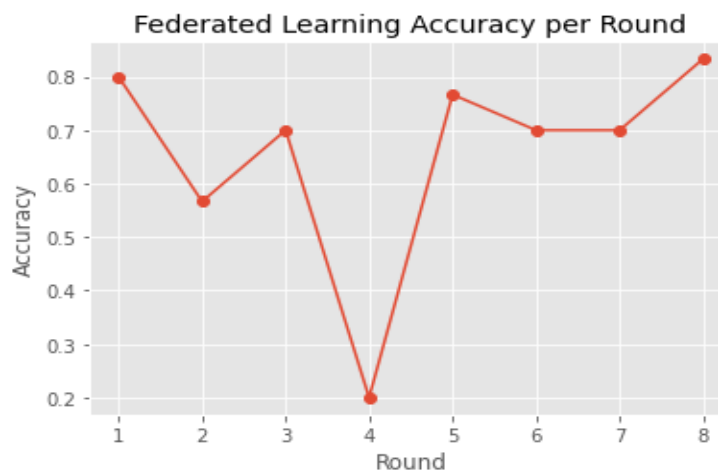


Figure 27: Federated learning accuracy per round

4.10 Social Network Analysis (SNA)

The social network analysis is an effective tool of conceptualizing the structure and dynamics of the social networks with the help of which it may be beneficial to the various fields to comprehend the flow of interaction, relations and communication. The main objective of this project is to apply SNA to the analysis of various types of social networks in various contexts including the analyses within organizational frameworks, community life, or communication trends in various digital platforms.

The primary goal of the SNA project would be to discover the relations between

individuals, groups, or entities in the network and learn how these relations can influence their behavior and decision-making methods in the world. Through the study of the interaction structure of nodes (people or objects), and edges (relationship) the study will aid in learning the different found patterns, clusters and influential nodes in the network. One should be able to comprehend the type of influence, identify the way of the communication manner, and examine the work of the network optimization.

In this respect, the project will involve the collection of data through various sources including surveys, social media interactions, organizational database, or forums. Methods used to examine such data are graph theory, centrality measures and community detection algorithms. An example is degree centrality, which indicates well-linked individuals in the network, and betweenness centrality, which provides a node with details of whether it is a tie bringing two disjunct groupings together. Joint clustering algorithms have been developed that determine communities or groupings in a network-the groups with which a node is connected into close ties.

Applications of this project are intersecting various fields, e.g. an organization development application wherein SNA investigates optimizing the patterns of communication and the leadership of communities and team networking. Modeling would also be applied by the field of public health in the interpretation of disease patterns of dissemination and individuals within a community that should be targeted through interventions. Lastly, in the case of digital marketing, SNA provides evidence of the fact that companies can exploit the influence of people who can sway the opinion of people about their products. The project will also demonstrate its contribution to research by reflecting a solid and valuable insight into academic disciplines regarding social dynamics, evolution of networks, and, in addition, the influence of external influences in altering the arrangement of a network.

This study illustrates the potential which analysis provides in social networks; in fact, it offers viable solutions to the optimization of relationships and communication in other fields. Whenever an individual employs such techniques, they are more intelligent regarding social interaction and therefore, enhances decision making, good policy principle, and strategic planning.

Network Property: It contains 965 nodes and 15, 259 edges: 965 nodes and 15,259 edges in a very densely intertwined system. The top five according to the degree centrality (65002, 268733, 2358969, 52130, 58873) are those that have a number of direct connections in the network hence they are heavy weights in the network. Then there is the top five nodes based

on the betweenness centrality.

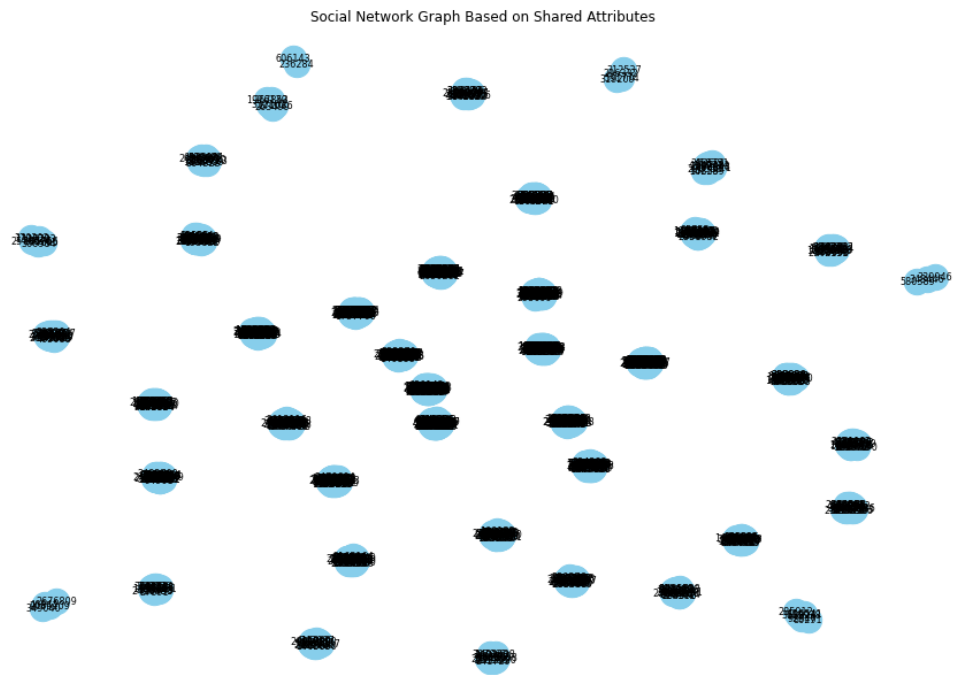


Figure 28: SNA graph based on attribute

Number of Communities Detected: 41

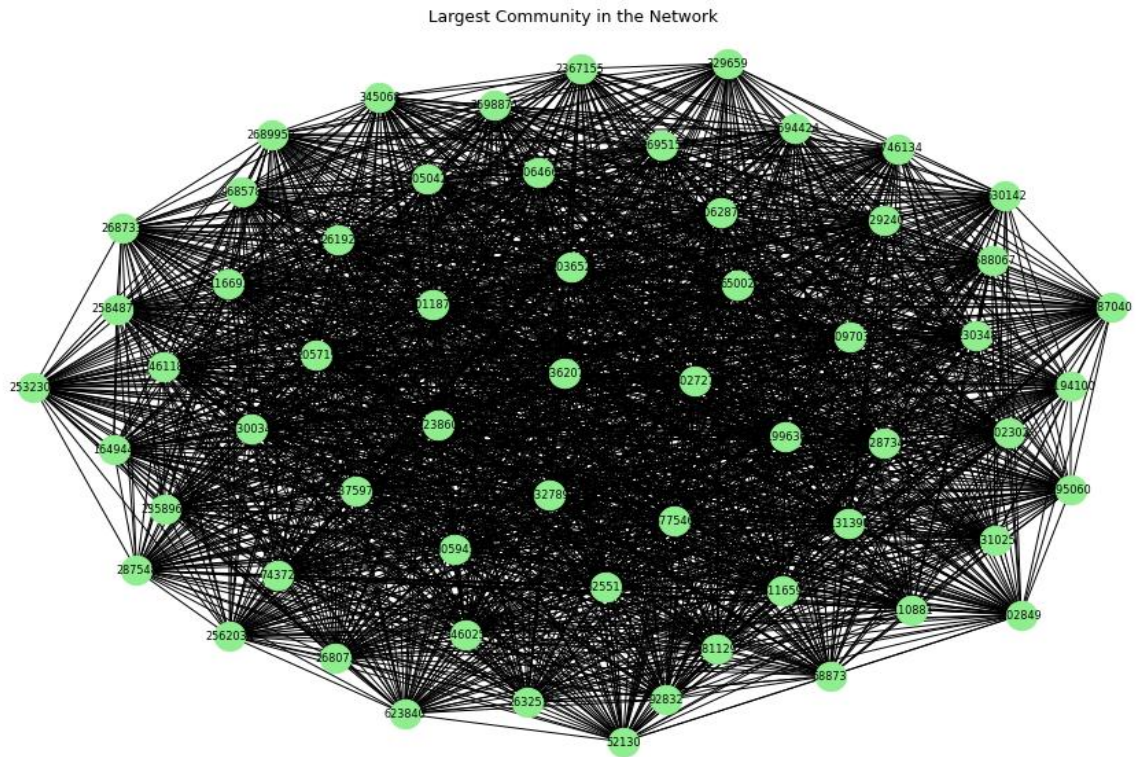


Figure 29: Community graph

Node Degree Distribution

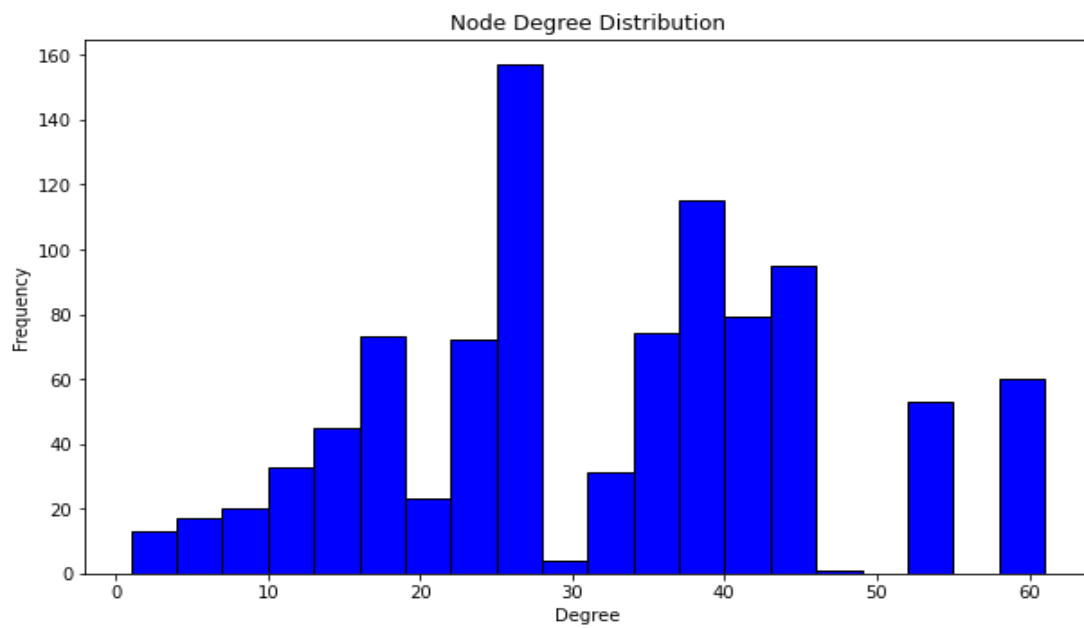


Figure 30: Node degree distribution

It is a histogram that demonstrates the concentration of a degree in a Social Network Analysis. In SNA, a node embodies individuals, organizations and financial accounts, and the links describe their connections or associations. The degree of a node is the amount of direct connections that it has. The x-axis of the plot represents the degree values, whereas the y-axis represents the frequency of frequency of that node which has that degree value. The degree distribution reveals that most nodes start with low degrees having few connections directly with each other, many nodes have extremely high degrees agglomerating many connections and serving as hubs or different influencers to the network. The pattern occurs either in a scale-free network or as a result of a power-law distribution, which is known to constitute a good fit to the real-life social networks. In these structures, most participants have few connections and a small group have so many connections that the small group becomes essential to the connectivity of the network. The fact that these nodes tend to become the major influencers or bottlenecks in a network makes the identification of these nodes with the high degrees quite important. Eliminating them would be very detrimental to the network structure.

Chapter 5 - Results and Discussion

5.1 Overview of Experimental Results

They report the findings of an experiment to estimate how different machine learning models could be used to predict student performance, as well as Federated Learning (FL)-based versions of them.

In the baseline models, Random Forest recorded the highest level of precision 91 percent with a high F1-score of 0.87 in macro-average classification. It displayed great accuracy and recall especially in the case of the category labeled as Pass. The elimination of those features, which included ``weighted_grade`` or ``pass_rate``, also resulted in decreases in accuracy (89 percent and 90 percent respectively) but in general performance was still high. XGBoost has shown a high accuracy of 90 percent with the pass category showing very good results, but the performance in both distinction and fail categories was also relatively lower with area precision and recall of 0.81-0.84.

Logistic Regression was seen to perform well with 88 percent accuracy and 0.83 F1-score with no features being taken out and there was accuracy reduction when some of the features were taken out. In contrast, LDA has a consistent accuracy of 88%, and better precision and recall in the Pass category and slightly lower scores in Fail.

In comparison, there was quite a high Level of variability on the Federated Learning-based models, with an accuracy range of 56% to 83.3% of accuracy. This variability is ascribed to conditions like variation in data distribution, IID representation of the data, and small amount of local training data. FL has the advantage of privacy since training can be decentralized regardless of the reduced accuracy.

5.2 Individual Model Performance Analysis

Logistic Regression is simple and interpretable model, and also the first choice when it comes to solving a binary-class problem. It performs effectively with relationships that are linear and small in feature sets but does not perform with non-linear data that are more complex. Although it is very simple, logistic regression is a reasonable baseline model which can be harmed by high-dimensional features or a high correlation of features.

Random Forest Classifier is deeply proficient in analyzing the importance of features and is capable of striking disparity in the dataset by averaging the findings of numerous decision trees. It produces strong resistance to overfitting and can be computationally demanding when working with large datasets. LDA Classifier is good provided that the data is linearly separable and it is interpretable. But it is weak at non-linear interaction. XGBoost has high accuracy, faster and superior performance capturing complex feature such as feature interaction hence, it is very useful to imbalanced data, but it has to be tuned to work in optimal conditions. CNNs excel at feature learning and in the case of image data, but fail to perform well against tabular. Finally, Federated Learning also has the privacy-protecting advantage to train models on decentralized data. It does not match centralized tools such as CNN and XGBoost in terms of their performance, but the privacy benefits it has offer make it useful in situations where data cannot be shared.

5.3 Cross-Model Comparative Insights

Listing the major differences between the centralized and federated models in terms of their performance and privacy levels is conducted. Such centralized approaches as XGBoost and CNNs are generally better (in accuracy and speed) compared to federated approaches. This occurs because all the data can be accessed within the same point, which enables the model to learn with all the data which further increases the predictability of the model. There is the increased potency of hardware and faster processing of data as well in centralized models. Conversely, Federated Learning (FL) gives rise to another property, which is privacy, which implies that it is possible to train models using decentralized data sources without exchanging raw data. Although in federated models a decrease in the accuracy is usually observed - as data availability is limited and likely to be non-IID (Independent and Identically Distributed), that compromise the effectiveness of centralized models, federated models offer a welcome alternative when the data are sensitive or privacy is of concern such as in healthcare and finance, applications.

Regarding the complexity of models, one will observe a trade-off between accuracy and the interpretability. Advanced models, e.g. the XGBoost and CNN, are associated with high accuracy but out of the expense of interpretability. Whereas their performance is superior in modeling complex relationships on data, they are less transparent as it is usually challenging to describe how their decision making process is based. Conversely, less complicated models such as Logistic Regression and LDA have easier to interpret results at the expense of

potentially inferior results on more complex data. The irreconcilability of accuracy and interpretation is paramount when it comes to choosing a model because stakeholders might want their model to be transparent whatever the effect on its functioning.

5.4 Privacy and Security Considerations

With the help of Federated Learning (FL), the security of student data is much less disastrous since they are never delivered to a cloud or server outside of the local device or institution. Compared to transferring raw data to a centralized server, FL is also used to facilitate local training of models with the data of each educational institution. The newer versions of the model are then disseminated in the form of aggregated parameters and the likelihood of being exposed to individual student data is greatly lowered. Such decentralized methodology can simplify meeting the requirements of data protection guidelines such as GDPR and FERPA since the data stays within institutional confines where the risk of leakage during the data transfer process is reduced.

FL offers an invaluable potential in multi-institution educational systems since they have the ability to mutually train on available Education data to them without any violation of the privacy of the student. As a case in point, higher institutions or schools located in other parts of the world can take their role in a unified framework that enhances the forecasting of student achievement, the detection of vulnerable students, or streamlining the learning processes without the transfer of sensitive student data across schools. Such a collaborative learning process improves the model performance, and more likely to establish the trust and transparency between the institutions involved, which is a requirement in sustaining the promotion of privacy and using shared data to more desirable educational outcomes.

5.5 Explainability Results

SHAP (SHapley Additive exPlanation) has been widely used in this study to provide some insights into explainability of each of the models and give a clear picture of the contribution of the features to the predictions. SHAP values can be used to see the impact each feature has on the decision of the model (like `weighted_grade`, or `pass_rate`) e.g. in Logistic Regression, SHAP displays how the interaction of the final result with those features not only influences the probability of passing or failing. The same can be said of the Random Forest and the XGBoost, where it is possible to find out, using SHAP values, the feature importance

and how features interact with each other, determining which aspects, such as `num_of_prev_attempts` or `pass_rate`, best predict success in students. In the case of CNN, SHAP is applicable to explain intricate learned features and their contribution to classification decisions, albeit less useful with respect to explaining the inner-working of feature extraction.

Regarding centralization vs federation, explainability would be the same with respect to the capability of SHAP to identify the importance of the feature, whereas the main distinction is the way to share model updates. The decision-making process of the model can be traced in every source of data available in a single location in a centralized arrangement. Nevertheless, within a federated environment, SHAP offers localized explainability across different local clients, which complicated the monitoring of global trends because pushing model updates is decentralized. Irrespective of this, federated learning still has (now lost) privacy but it cannot lose interpretability, serving as an understanding of the local behavior of the models but in confidentiality.

Chapter 6 - Conclusion and Future Work

6.1 Summary of Objectives and Achievements

The main aim of the study was to generate and test different machine learning models to predict the outcome of the student performance, and especially their interpretability and ability to preserve privacy. The research problem sought to compare the models in terms of their effectiveness on predicting an outcome like pass, fail or distinction based on features passed in the form of weighted_grade, pass rate and num of prev attempts.

The overall goal was attained through the contributions of each of the models. The simple prediction was provided by Logistic Regression, giving evidence of its capabilities to evaluate the connection between the primary characteristics and the success of the students. Random Forest Classifier worked best with imbalance data and was used to achieve good inference regarding feature significance, but XGBoost added low error rate and speed, which was a sign of complex relationship between features. LDA was useful in analyzing data that can be separated linearly and that would appear better in the interpretation, but CNN demonstrated its capabilities when engaging in hierarchical features and failed when faced with tabular data. Finally, Federated Learning (FL) was also added to evaluate how much the concept of decentralized learning could affect the privacy, showing that many institutions could collaborate to train their models without revealing sensitive data of the students, another step towards the system secrecy.

Overall, all the models did their respective job, and Federated Learning could present an extra layer of privacy protection and demonstrate a high level of prediction accuracy too, making it a tangible option in the real-life educational system.

6.2 Key Findings

The principal conclusions of the work are that XGBoost has been found to be the most accurate model overall with remarkable results of 90 percent accuracy in predicting the outcomes of students. This was because it could manage features interactions especially between the weighted grades and pass rates which gave it the capacity to clearly differentiate between passing, failing, and distinction categories. The result of the model was not sensitive to the different configurations, and it was consistent with the high score that shows that the model is robust or the prediction of student performance was not likely to be affected much

in the absence of some features.

In the comparison of Federated Learning (FL) with the centralized model, it has been noted that XGBoost and CNN models perform very well in the centralized scenarios; however, FL had a trade-off in terms of accuracy versus privacy. The slight decrease in the prediction accuracy could be explained by the Federated Learning approach to data privacy that implies the local training of a model and the exchange of updated models. This owes to the difficulties of non-IID (Independent and Identically Distributed) data and low communication rounds in FL which may retard convergence. Nevertheless, with the privacy advantages of FL, it may be a very strong option in contexts where student data security is of paramount importance and especially in multi-institutional or cross-institutional applications.

Altogether, XGBoost was the most accurate model; but Federated Learning has been valuable since it balances between accuracy and privacy, a dimension that places it as a potential model in educational systems that need to use data with the protection of privacy and the maintenance of its practical properties.

6.3 Limitations of the Study

Although the method of this study is very in-depth, there are a number of constraints concerning hardware and training time. The hardware limitation was one of the main problems since it affected training more complicated models, such as CNN and XGBoost. Their models were very heavy in terms of computation power hence, taking a long time to train. This has meant that fewer iterations were possible on tuning of hyperparameters, a scenario that may have jeopardized the maximization of fully optimizing the models, a factor that might have compromised the performance.

Furthermore, additional complexity and training time was introduced when Federated Learning (FL) was used. FL contains many clients that train conditionally and this can slow down the convergence process as there are some model weight aggregation and communication overhead there that need to occur on a frequent basis. The non-homogenous data distributions among clients also provided complexity to the training process because it needed to be highly synchronous to encourage the use of the identical model update. In cases where privacy was beneficial by FL, it also resulted in slower model training as compared to centralized schemes. These disadvantages mean that the additional research may capitalize on better hardware and effective FL algorithms to speed up.

6.4 Future Research Directions

The enhancement of the performance, privacy and the applicability aspect of the research is some of the sophisticated approaches that can be employed in future predictive modeling studies in education. One of the possible paths that might be successful is the integration of applications of differential privacy and secure multi-party computation (SMPC) into federated learning. Each of these methods will ensure the safety of confidential student information, but will still allow decentralized information training. The employment of these privacy saving techniques will reduce the problem of security and make sure that the predictions are correct.

Furthermore, it has a high opportunity to widen the spectrum of predictive models to look at additional major educational requirements, i.e., to predict dropout, personalized learning plans, and engagement forecasting. The applications are in a position to provide certain interventions, facilitate course design and improved resource use in learning establishments.

Furthermore, as state of art deep neural nets like Transformer models and Reinforcement Learning (RL) get more integrated, the accuracy of the predictions regarding education may be significantly improved. Successful transformer models in natural language processing (NLP) might be able to learn more varied student learning behaviors, and Reinforcement Learning might allow the possibility that personalized learning systems were dynamically adaptable enough to satisfy individual student needs, with time yielding a learning system that would be optimized. Such advances in federated learning (as well as deep learning) are soon poised to transform the field of educational analytics and personalised learning.

6.5 Final Remarks

The other significant advancement that has enormous potential in the education field is the integration of Machine Learning (ML) and Deep Learning (DL) and privacy-preserving techniques, such as Federated Learning (FL). As interest in using data-driven solutions to forecast a student performance and, consequently, enhance it keeps increasing, it is crucial to balance the precision of the model with the required extent of data privacy. Models of learning, including XGBoost, CNN, and Random Forest offer unprecedented prediction and the opportunity to discover the most complex patterns in student data. However, the use of data of personal and sensitive student information is involved, and it leads to privacy and security concerns primarily in a multi-institutional context.

Federated Learning make such considerations and has the option of training models by training local devices without the transmission of sensitive information. This practice as well renders the information of students secure and confidential and can still offer institutions the most treasured predictive models. By combining these state-of-the-art models in conjunction with privacy-saving solutions, institutions may be able to transform the potential of AI into tangible learning results with the added advantage of maintaining the privacy of student data.

Besides, the specified approach is particularly timely because of a tendency to transform the educational systems into at least slightly less centralized and cooperative systems. Future could be the creation of systems that will be capable of providing usable information concerning the way students are doing and at the same time safeguard their privacy. Undoubtedly, as privacy-preserving ML/DL models continue to evolve, they will trigger new doors concerning personalized learning experiences and predictive analytics and adaptive learning systems to allow education to be more accessible, productive, and secure to all students.

References

- [1] K. Fahd, S. Venkatraman, S. J. Miah, and K. Ahmed, ‘Application of machine learning in higher education to assess student academic performance, at-risk, and attrition: A meta-analysis of literature’, *Educ Inf Technol (Dordr)*, vol. 27, no. 3, pp. 3743–3775, Apr. 2022, doi: 10.1007/S10639-021-10741-7/METRICS.
- [2] K. M. Hasib, F. Rahman, R. Hasnat, and M. G. R. Alam, ‘A Machine Learning and Explainable AI Approach for Predicting Secondary School Student Performance’, *2022 IEEE 12th Annual Computing and Communication Workshop and Conference, CCWC 2022*, pp. 399–405, 2022, doi: 10.1109/CCWC54503.2022.9720806.
- [3] G. Ben Brahim, ‘Predicting Student Performance from Online Engagement Activities Using Novel Statistical Features’, *Arab J Sci Eng*, vol. 47, no. 8, pp. 10225–10243, Aug. 2022, doi: 10.1007/S13369-021-06548-W/FIGURES/8.
- [4] Z. Wu, L. Huang, Q. Huang, C. Huang, and Y. Tang, ‘SGKT: Session graph-based knowledge tracing for student performance prediction’, *Expert Syst Appl*, vol. 206, p. 117681, Nov. 2022, doi: 10.1016/J.ESWA.2022.117681.
- [5] D. A. Shafiq, M. Marjani, R. A. A. Habeeb, and D. Asirvatham, ‘Student Retention Using Educational Data Mining and Predictive Analytics: A Systematic Literature Review’, *IEEE Access*, vol. 10, pp. 72480–72503, 2022, doi: 10.1109/ACCESS.2022.3188767.
- [6] W. Xiao, P. Ji, and J. Hu, ‘A survey on educational data mining methods used for predicting students’ performance’, *Engineering Reports*, vol. 4, no. 5, p. e12482, May 2022, doi: 10.1002/ENG2.12482.
- [7] S. S. Shreem, H. Turabieh, S. Al Azwari, and F. Baothman, ‘Enhanced binary genetic algorithm as a feature selection to predict student performance’, *Soft comput*, vol. 26, no. 4, pp. 1811–1823, Feb. 2022, doi: 10.1007/S00500-021-06424-7/METRICS.
- [8] A. Kukkar, R. Mohana, A. Sharma, and A. Nayyar, ‘Prediction of student academic performance based on their emotional wellbeing and interaction on various e-learning platforms’, *Educ Inf Technol (Dordr)*, vol. 28, no. 8, pp. 9655–9684, Aug. 2023, doi: 10.1007/S10639-022-11573-9/METRICS.
- [9] M. N. Yakubu and A. M. Abubakar, ‘Applying machine learning approach to predict students’ performance in higher educational institutions’, *Kybernetes*, vol. 51, no. 2, pp. 916–934, Feb. 2022, doi: 10.1108/K-12-2020-0865/FULL/XML.
- [10] G. Deeva, J. De Smedt, C. Saint-Pierre, R. Weber, and J. De Weerd, ‘Predicting student

- performance using sequence classification with time-based windows’, *Expert Syst Appl*, vol. 209, p. 118182, Dec. 2022, doi: 10.1016/J.ESWA.2022.118182.
- [11] A. Asselman, M. Khaldi, and S. Aammou, ‘Enhancing the prediction of student performance based on the machine learning XGBoost algorithm’, *Interactive Learning Environments*, vol. 31, no. 6, pp. 3360–3379, Aug. 2023, doi: 10.1080/10494820.2021.1928235.
- [12] Y. A. Alsariera, Y. Baashar, G. Alkaws, A. Mustafa, A. A. Alkahtani, and N. Ali, ‘Assessment and Evaluation of Different Machine Learning Algorithms for Predicting Student Performance’, *Comput Intell Neurosci*, vol. 2022, no. 1, p. 4151487, Jan. 2022, doi: 10.1155/2022/4151487.
- [13] S. Poudyal, M. J. Mohammadi-Aragh, and J. E. Ball, ‘Prediction of Student Academic Performance Using a Hybrid 2D CNN Model’, *Electronics 2022, Vol. 11, Page 1005*, vol. 11, no. 7, p. 1005, Mar. 2022, doi: 10.3390/ELECTRONICS11071005.
- [14] G. Feng, M. Fan, and Y. Chen, ‘Analysis and Prediction of Students’ Academic Performance Based on Educational Data Mining’, *IEEE Access*, vol. 10, pp. 19558–19571, 2022, doi: 10.1109/ACCESS.2022.3151652.
- [15] S. Hussain and M. Q. Khan, ‘Student-Performer: Predicting Students’ Academic Performance at Secondary and Intermediate Level Using Machine Learning’, *Annals of Data Science*, vol. 10, no. 3, pp. 637–655, Jun. 2023, doi: 10.1007/S40745-021-00341-0/FIGURES/4.
- [16] S. Batool, J. Rashid, M. W. Nisar, J. Kim, H. Y. Kwon, and A. Hussain, ‘Educational data mining to predict students’ academic performance: A survey study’, *Education and Information Technologies 2022 28:1*, vol. 28, no. 1, pp. 905–971, Jul. 2022, doi: 10.1007/S10639-022-11152-Y.
- [17] H. Pallathadka, A. Wenda, E. Ramirez-Asís, M. Asís-López, J. Flores-Albornoz, and K. Phasinam, ‘Classification and prediction of student performance data using various machine learning algorithms’, *Mater Today Proc*, vol. 80, pp. 3782–3785, Jan. 2023, doi: 10.1016/J.MATPR.2021.07.382.
- [18] Y. Chen and L. Zhai, ‘A comparative study on student performance prediction using machine learning’, *Educ Inf Technol (Dordr)*, vol. 28, no. 9, pp. 12039–12057, Sep. 2023, doi: 10.1007/S10639-023-11672-1/METRICS.